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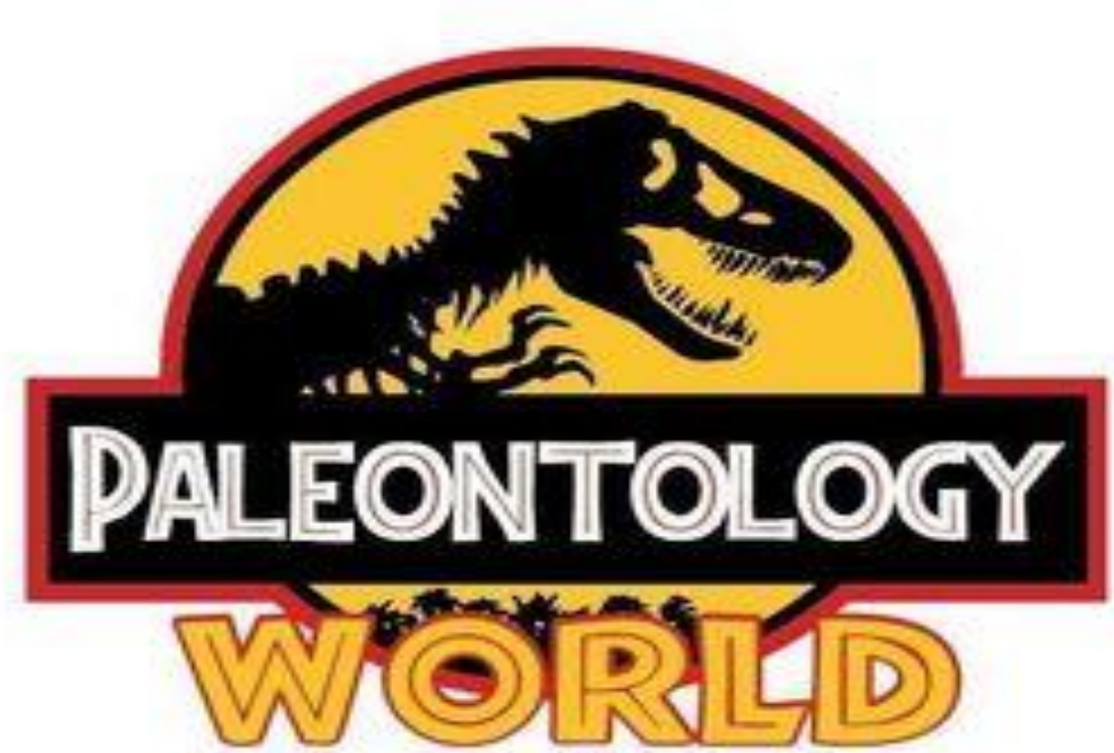
PAST PAPERS /MCQs/ MID TERM /FINAL TERM/SOLVED QUESTIONS

MID TERM & FINAL TERM

QUIZ NO 1 | QUIZ NO 2 | GRAND QUIZ | MID TERM | FINAL TERM

MSC-ZOOLOGY

ZOO503-Zoogeography-and-Paleontology



PAST PAPERS

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QUIZ NO AND MID TERM MCQs

1. Which of the following is considered as health threatening microorganism of water?_____ _____ _____**Vancomycin resistant Enterococus spp**
2. The first vertebrates fish had no _____. Fossil sites at Chengjiang in China: Early Cambrian?_____ **Jaws**
3. There are _____ or so living echinoderm species include familiar forms such as sea lilies, sea urchins, sand dollars, starfish and sea cucumbers_____**6000**
4. The Ediacaran biota consisted of _____ and frond-shaped?_____ **Enigmatic tubular**
5. Diatoms are _____ autotrophs?_____ **Unicellular**
6. The evolution of fishes start from the _____ to the present day, and the Paleozoic tetrapods?_ _____ **Cambrian**
7. When a fish became a land animal, surely the key problem was _____? **Breathing air**
8. Skeletons that are made from several parts may become ....., separated into their component parts._____ **disarticulated**
9. _____ is the term in which the males and females show different sizes, and different specialized features._____ **Sexual dimorphism**
10. The _____ dominated the filter-feeding benthos of the Paleozoic ?_____ **Brachiopods**
11. "imperfection of the geological record" is pointed out by_____ **Charles Darwin**
12. The first pennate diatoms appeared during Paleogene colonizing freshwater and this group reached an acme during the _____. **Miocene**
13. Incomplete remains of ..... were discovered in 1856_____ **Neandertal man**
14. __ are the organic carbons much less liable to break down, such as cellulose._**Refractories**
15. The Sub phylum Crustacea Only ___ groups, have significant geological records?_____ **Two**
16. In porifera, _ have a number of simple chambers with a single central paragaster?___ **Sycon**
17. Bacteria in rocks up to _____ Ga, indicated by stromatolites._____**3.2**
18. In Cnidarians, the stage which is free and swims are called?_____ **Medusae**
19. ____ are considered as rare earth elements._____ _____**all**
20. The best studied embryo fossil of metazoans are Doushantuo formation?_____ **true**
21. Father of comparative anatomy is _____**Cuvier**
22. The ____ are filter-feeding members of the sessile benthos?_____ **Sponges**
23. Earth has oscillated at least ____ times between icehouse and greenhouse conditions.- ____**5**
24. Insects had probably already evolved flight before the _____, when giant dragonflies patrolled the forests._____ **Mid Carboniferous**
25. The Brachiopoda, diversified at _____ era?_____ **Paleozoic**
26. Geographic range _____ may be large or small, and controlled by a variety of factors including _____._____ **Both**
27. The class insecta of hexopods have _____limbs on their body?_____**6**
28. Which type of diatoms preserved in fossil record?_____ **A succession of contiguous**
29. formations with common characteristics is called a ____._____ **group**
30. Physical and chemical effects occurring after burial, are termed as ____. _____**diagenesis**
31. Extinction and dating rocks is the use of ____._____ **paleontology**
32. The similarity of a death assemblage to its living counterpart, is its _____. **Fidelity**
33. Important factor to drive evolutionary change_____ **Geographic change**
34. The use of ecometerials on walls is effective for?_____ _____**Sound suppression**
35. A compound…. isolated from neam plant has broad spectrum insecticidal effect___ _____ _____**Azadirechtin**
36. Which of the following Phytoplankton facilitate the growth of fishes?_ __**Chlorella vulgaris**
37. Which of the following chemicals is involved in the detoxification of the metals?_____ _____ **cu-EDTA complex**
38. Main purpose of growing non food crops is to get__ commercially?_____ _____**Oils**
39. Which of the following is used to degrade the xenobiotics___ __**Mixed microbial cultures**
40. Which of the following has been considered as a strong anti-biofouling agent?_ ___**Acid rain**

41. Zooplanktons are microscopic .....?_____Animals
42. The _____ is the oldest Cnidarian?_____ Chondrophora
43. At which pH, air contamination is controlled? _____7.5
44. The Phylum Mollusca bear a body plan based on _____ features?_____ Four
45. volatiles, forms of carbon in formation of …….parts of body animals _____ soft
46. the wide hinged fossils are _____Spiriferide brachiopods
47. Early mineralization of soft tissues depends on …factors_____3
48. Specimen displayed at the Museum für Naturkunde in _____Berlin
49. Scottish scientist ---------- (1726-1797) challenged no vestige of beginning no prospect of an end _____James Hutton
50. The earth today can be divided into ---------- main provinces of biogeography_____ Six
51. Paleontology is study of _____old creatures
52. The _____ lack gills and eyes, have a mouth equipped with a radula and foot surrounded by tentacles similar to that of the bivalves, adapted for burrowing?_ _____Scaphopods
53. The Class Gastropods evolved in _____? _____Entire Phanerozoic
54. "All men are mortal. Socrates is a man. Therefore Socrates is mortal."Who said this?_____ _____ Aristotle
55. The _____ is an integral part of the body plan of an animal, providing support, protection and attachment for muscles_____ Skeleton
56. _____ were involved in the generation of molecular oxygen._____ Cyanobacteria
57. The Insect diversified in freshwater during the Late _____?_____ Silurian
58. By the Late _____ a very diverse insect fauna had evolved, with forms such as the dragonflies and mayflies capable of powered flight?_____ Carboniferous
59. The metazoans having arthropod–nematode–priapulid plus other taxa group are known as?_____ Ecdysozoans
60. In arthropods, skeleton growth by periodic bursts between intervals of ecdysis or molting?____ _____ True
61. Pasteur performed an experiment in term of Spontaneous generation in _____. 1861
62. …….and……Recognized the reality of biogeographic provinces. _____ both
63. fossil can be used as _____Ornaments
64. Following are examples of biological species concept_____ all
65. Paleozoic describes the ____ life._____ ancient
66. Humans arose _____ Ma, and fossil evidence points to repeated human migrations out of Africa?_____ 6-8
67. The first metazoans were _____ with one main cell type & peripheral collar cells or choanocytes, with a whip or flagellum_____ Multicellular
68. The _____ achieved great diversity and abundance only after the extinction of the dinosaurs?_____ Mammals
69. The investigation of thermal maturation is now a routine_____ _____ _____ _ petroleum exploration technique
70. The evolutionary radiation of animal life during the _____ Period, 40 my after the CE?_____ Ordovician
71. If there is no change in proportions during growth, the feature is said to show ____ growth.____ _____isometric
72. the first soft-bodied Ediacaran fossils were found in _____Australia
73. Slight genetic change over a few generations in a population that does not result in the evolution of a new species is called as_____. _____Micro evolution
74. Quantitative techniques to understand the relationships between ---------- thickness and time. And to make link from locality to _____Rock
75. Chemostratigraphy is a type of _____ Stratigraphy
76. Paleontologists should be ….when they attempt to reconstruct a whole plant or animal, and try to understand its biomechanics_____ careful

77. Fossils information is also useful in _____ **Geology**
78. ……was linked to the idea of evolution_____ **Progressionism**
79. Tectonic units, such as thrust sheets _____ **tectonostratigraphy**
80. ….is a branch of geology that attempts to subdivide and link sedimentary deposits into unconformity bound units on a variety of scales_____ **Sequence Stratigraphy**
81. ……Allows the tectonic history of mountain ranges to be reconstructed_____ **Fossils**
82. was an antiprogressionist_____ **Charles Lyell**
83. There are _____ main groups of conodonts?_____ **3**
84. The German scientist ---------- suggested that the continents moved across the earth surface on a liquid core, suggesting that continent could in fact drift_____ ____ **Alfred Wegener**
85. Fossil sharks' teeth were known as _____ **glossopetrae**
86. According to ____, common patterns would point to an explanation, or law of nature._____ _____ **Sir Francis Bacon**

Grand quiz

1. The osteichthyans or bony fishes also appeared in the_____ _____ **Devonian**
2. In porifera ---------- have sacs with a single chamber lined by flagellate cells_____ **Ascon**
3. Which of the following is the second major diapsid clade _____ **Lepidosaurs**
4. The best studied embryo fossils of metazoan are doushantuo formation from ____ **South china**
5. According to -------- common pattern would point to an explanation or law of nature _____ _____ **Sir Francis bacon**
6. The geographic model based on the establishment of geographic _____ **Barriers**
7. The --------- is the morphologically diverse clade of protostome animals _____ **The spiralians**
8. The armored fishes were abundant in -------- seas and lakes _____ **Devonian**
9. The metazoans having echinoderm hemichordate- chordate group are known as _____ **Deuterostomes**
10. The Cambrian explosion have relatively short span event of ------my?_____ **20**
11. One of the most remarkable echinoderm animal ---------- occur in the upper Ordovician succession of the Craighead inlier,north of the Girvan valley , southwest Scotland____ **Star fish**
12. Radiolarians are --------- like protozoan with siliceous tests_____ **Animal**
13. Condition that are necessary for the survival of hard parts in the sediments_____ __ _____ **Chemical condition**
14. Quantitative techniques to understand the relationships between ---------- thickness and time. And to make link from locality to locality._____ **Rock**
15. The class insect of hexapod have ---------- limbs on their body_____ **6**
16. The earth today can be divided into ---------- main province of biogeography_____ **Six**
17. --------- are the beasts that live in or the seabed_____ _____ **Benthos**
18. The trilobites extinct at the ---------- time?_____ _____ **End Permian**
19. The class cephalopods have ---------- subclass _____ **Three**
20. The mites, scorpion and spiders belong to the subphylum____ _____ **Chelicerata**
21. If there is no change in properties during growth, the feature is said to show ---------- growth _____ **Isometric**
22. proof is impossible said_____ _____ **Karl popper**
23. The _____ diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels._____ **Ordovician**
24. Fossils form tell us about_____ **Adaptation through time**
25. In which year did Darwin wrote the book "On the Origin of Species" _____? _____ **1859**

26. All organisms were linked in time by a unidirectional ladder, from ____ at the bottom to most ____ at the top._____ **simplest to complex**

27. Paleontologists should be ..... when they attempt to reconstruct a whole plant or animal, and try to understand its biomechanics, when they have just a few bones or bits of twigs____ **careful**

28. Infaunal organisms live _____ the sediment–water interface._____ **beneath**

29. The Subphylum Crustacea, class Malacostracans have _____ subclasses?_____ **Two**

30. In end of Ordovician, event like glaciation & loss of almost cause _____ % of brachiopod families?_____ **80**

31. "Great Chain of Being" is the idea of _____. **Lamarck**

32. ---------- is and understanding of geological time_____ _____**Stratigraphy**

33. The ediacara biota, first impression of soft bodies organism of upper Proterozoic was found in_____ _____**Namibia**

34. Early mollusks first rises at ---------- age _____**Early Cambrian**

35. The group may prove to have as many as 10 million living _____**Hexapods**

36. species when the rich founas of the tropics have completely _____**described**

37. Chemical processes of breakage include ____._____ **all**

38. The ---------- were a sister group to the mollusk _____**Annelid worms**

39. Average temperature in humid tropical zone is ------- degree _____**Above 18**

40. Insect had probably already evolved fight before the ---------- when gain dragonflies patrolled the forests____ _____**Mid – carboniferous**

41. The first pinnate diatoms appeared during Paleogene colonization freshwater and this group reached an acme during the _____**Miocene**

42. The idea of plastic forces had been largely overthrown by the _____**1720s**

43. The principle of cross cutting relationship _____**If a body or discontinuity cuts across a stratum it must have formed after the stratum**

44. ---------- in which males and females show different size, and different specialized feature _____**Sexual dimorphism**

45. Bivalve was first named by ---------- in 1758? _____**Linnaeus**

46. The ---------- was the earliest arthropods in geological time scale _____**Trilobites**

47. In diploblastic body plan having outer ectoderm and inner ectoderm. These two layers are separated by _____**A cellular mesogloes**

48. A succession of contagious formation with common characteristics is called _____**Group**

49. The ---------- is the recent cnidarian _____**Ediacaran**

50. First mollusks descendent from like living _____**Flatworms**

51. Humans are most closely related to _____**Chimpanzee**

52. Population growth and resources availability are ---------- to each other _____ _____ **Inversely proportional**

53. The ---------- were heavily hit by the end-Permian mass extinction event and most notably the dinosaurs were key forms through the Mesozoic _____**Reptiles**

54. If there is no change is proportions during growth, the feature is said to show___ _**Isometric**

55. Sphenodon the tuatara of ---------- _____**New Zealand**

56. Stratigraphic ranges can also be influenced by the _____**Signor- lips effect**

57. Track way from middle Ordovician rock in the English lake district suggest that arthropods invade land about _____**50**

58. It is believed that the center of the archaeocyatha origin are now located in __**East Siberia**

59. Human baby tail and excessive hairs related to term _____**Atavisms**

60. The parazoan body plan characterized by cells organized in two __**Wandering cells or**

61. layers separated by jelly like material punctuated by so called ---- _____**ameobocytes**

62. Wings of birds related to the term _____ **Homology**

63. Precise biotic and morphological changes along phylogenetic work of john Phillips _____ _____ **The linage**

64. Human arose -------- ma, and fossil evidence point to repeated human migration out of Africa_____ **6-8**

65. The first vertebrate fish had no ---------- fossils site at changing in china early Cambrian _____ **Jaws**

66. The subphylum myriads first appeared during the _____ **Mid Silurian**

67. The ---------- diversification generated witnessed a staggering increased in biodiversity at the family , genus and species level _____ **Ordovician**

68. ---------- means the rock has been covered by younger rock _____ **Burial**

69. The ---------- is common in the fossils record from the Ordovician through the Devonian. Especially abundant in the Silurian and Devonian _____ **Stromatoporoidea**

70. Paleontology lies on the border _____ **b/w biology and Geology**

71. Species help shape the ecosystem and that can trigger large scale changes if the disappeared _____ **Keystone**

72. A classic keystone species is the ---------- _____ **Elephant**

73. The function of single fossil organisms ---------- _____ **Paleoecology**

74. Best amount of fossils that are present at sediments water interface _____-_____ _____ **Both epifaunal and infaunal organisms**

75. The successfully built reefs in unstable environments _____ **Archaeocyaths**

76. By the late carboniferous a very diverse ---------- fauna had evolved with forms such as the dragonflies and mayflies capable of powered fight? _____ **Insect**

77. What is the living status of graptolites which belong to phylum hemichordate_____ **Extinct**

78. Trackways from Middle Ordovician rocks in the English Lake District (Johnson et al. 1994) suggests that arthropods invade land about ____ myr earlier? _____ **50**

79. The _____ group may prove to have as many as 10 million living species when the rich faunas of the tropics have been completely described?_____ **Hexapods**

80. The partial or complete remains of plants or animals are considered as ____ fossils.___ **body**

81. Which Class occupy a pivotal position in Molluskan evolution?_____ **Rostroconchs**

82. Cellular fossils ___ Ga are highly controversial._____ **3.5**

83. The Subphylum Crustacea first appeared in the _____? **Cambrian**

84. The cohesion of a species is maintained over its natural range by processes of _. **gene flow**

85. The _____ is common in the fossil record from the Ordovician through the Devonian. Especially abundant in the Silurian and Devonian ? _____ **Stromatoporoidea**

86. Archaeopteryx", true "missing link" predicted by ___. _____ **Darwin**

87. Life on earth has evolving for up to --------- billion years_____ **4**

88. The oldest known amniotes hylonomus from the mid ------- of Canada_____ **Carboniferous**

89. The _____ phylum is least complex of the true metazoans (eumetazoans)___ **Cnidarians**

90. The _____ are metazoans composed of multicellular complexes with few cell types and lack variation in tissue or organs i.e. The sponges ?_____ **Parazoans**

91. The brachiopods experienced _____ main extinction events?_____ **Five**

92. ..... more complex, with cones, bars and blades._____ **Euconodonts**

93. Infaunal organism live ---------- the sediment- water interface _____ **Beneath**

94. Material forming any stratum were continuous over surface of the earth unless some other solid bodies stood in the way is the --------- of lateral continuity _____ **Principle**

95. The oldest eukaryote fossils may be _____ **Grypania**

96. The ---------- mass extinction was the largest of all time _____-**End- Permian**

97. Function of RNA is ---------- _____Protein synthesis
98. The astonishing tails and colour of male birds is an example of _____Breeding
99. ---------- fossils comprise of small samples , drill cores and chippings _____Microfossil
100. Condition that are necessary for the survival of hard part in sediments _____ _____ ___ Chemical condition
101. The oldest jaw bearing fishes were the placoderms such as _____ _____Coccestus
102. The ---------- are called belly- footed mollusks _____Gastropods
103. The phylum -------- are usually unsegmanted soft bodies animals with a body plan based on four features _____ Mollusk
104. The bryozoans have ---------- living species found today?_____6000
105. The functions of single fossil organisms _____ paleoautecology
106. Protists have _____ mode of nutrition._____ Autotrophic
107. _____ is One of the oldest multicellular organism and a member of a modern group that engages in sexual reproduction._____ Bangiomorpha
108. ---------- were like to idea of evolution _____Progressionism
109. Mainly the trilobites and over ---------- species are known _____15000
110. develop a new more objective method in systemic which has come to be called cladistics _____ Hennig
111. in June 1980 km meteorite(asteroid) had hit the earth _____10
112. The centrals are type of diatoms prefers ---------- _____Marine water environment
113. Following are example of biological concept _____All of above
114. In bryozons all the species live as _____Colonial
115. Unicellular microbes lack ----------and---------- include the carbonate producing cyanobacteria organelles) _____Both(nuclei and organelles
116. Cech called his discovery a ----------_____Ribozyme
117. Who gave the hypothesis that Earth is a living organism in equilibrium with its biosphere _____? _____Gaia
118. The _____ are called belly-footed mollusks?_____ Gastropods
119. The order and succession of rocks is called _____._____ Lithostratigraphy
120. The Phanerozoic and the current extinction crises is sometime 6th called ---------- _____extinction
121. The fossils record is ---------- _____Rarely complete
122. The ---------- have unique group I its own right unique porous structure and cellular body plan and lack true tissue? _____ Porifera
123. Population and paleocommunities may be analyzed with a range of ---------- techniques _____ Statistical
124. Addition of caco3 in burial called ---------- _____Carbon concentration
125. The _____ have unique group in its own right, unique porous structure and cellular body plan and lack true tissues?_____ _____Porifera
126. Right order of the coenoses is __Biocoenosis> thanetocoenosis> Taphocoenosis
127. In Deccan traps in ---------- represent a vast outer poring of larvae that courred over 2-3myr spanning the kt boundary _____ India
128. in ---------- evolution is when feature or organism evolve to look the same perhaps because they live the same way _____ Convergance
129. Shells. bones and wood may undergo -_____bioerosion
130. The arachnids of Arthropods: mainly comprise the terrestrial and scorpions?_____Spiders
131. The Brachiopoda. first appeared at -_____Early cambrian
132. The Class ….have well-defined head with the foot modified into tentacles?___Cephalopods

133. During mass extinctions, percent of species were wiped out._____**20-90%**

134. The Phanerozoic, and the current extinction crisis is sometimes called the extinction._____**6th**

135. The exoskeleton of arthropods is made up of 2_____**Chitin**

136. The individuals of Phylum arthropods often fused head and thorax which is called _____ _____**Cephalothorax**

137. …..in which males and females may show different sizes. and different specialized features._____ _____**sexual dimorphism**

138. The animal of phylum… differentiated into a head. thorax and abdomen?_____**Arthropods**

139. Brownish Sludge comprises of_____ _____**all**

140. Rocks and sand left behind by an advancing glacier is known as_____**tilite**

141. The theory of evolution by natural selection developed in -_____**1830s**

142. In Bryozoans. all the species live as ?_____**Colonial**

143. ….is being used for studying origin of life._____ _____**Paleontology**

144. Bivalves were first named by……in 1758?_____**Linnaeus**

145. The oldest locomotory trace fossils of metazoans are from about 550 Ma northwest a_____ _____**Russia**

146. Body plans of metazoans are usually defined by the number and type of enveloping walls of tissue together with the presence or absence of a _____?_____ **Celom**

147. All species today had evolved from other species in the past is related to the idea of ___. _____**common decent**

148. When microbes, such as fungi or bacteria, transform tissues of the dead organism, the process is termed ___._____ **decay**

149. Bacteria and Archaea are ;_____**prokaryotes**

150. Most famous developmental genes are the _____ genes_____ **homeobox**

151. The membrane which encloses the embryo is called_____ _____**the amnion**

152. The Phylum _____ consist of sea urchins, starfish & sea cucumbers ?____**Echinoderms**

153. The is the Ancient cups?_____ _____**Archaeiocyatha**

154. …..is a method of reconstructing phylogeny based on the identification of shared derived characters.___ _____**Cladistics**

155. ____ portrayed a clear sequence of laterally continuous, horizontal strata showing the concept of superposition.___ _____ **da Vinc**

156. There are _____ class in the Phylum Echinoderms?_____**6**

157. Early bony fishes almost certainly had_____ _____**both lungs and gills**

158. The first shark like chondrichthyans or cartilaginous fishes, came on the scene during the Early _____?_____ **Devonian**

159. The oldest rocks are from_____**canada**

160. The first widely accepted cellular fossils date from __Ga_____**2.5**

161. The Class Rostroconchs of Mollusks have genera?_____**35**

162. The branch of geology concerned with establishing the absolute ages of strata is termed as_____**Chronostratigraphy**

163. Chronostratigraphy is the branch of geology that is concerned with establishing the absolute ages of _____?_____ **Strata**

164. are the building blocks of communities_____ **Populations**

165. ….of strata is fundamental to all stratigraphic studies._____ ____**Superposition**

166. Apomorphies are features that arose ___ in evolution._____ **once**

167. In natural sciences, there have been ___ main approaches._____**2**

168. 'Ontogeny recapitulates phylogeny' is a law of ____. _____**Heckle**

169. The Trochophore larva type found in? _____**Mollusks and the polychaetes**

170. The Sub phylum Crustacea Only groups, have significant geological records?_____**Two**

171. The two groups of parazoan, the …and ….form the basal parts of the metazoan tree, diverging during Neoproterozoic?___ _____**Porifera and Cnideria**

172. The Nauplius larva is found in?_____ _____**crustaceans**

173. The transit of Indie from Gondwana to Asia is an Example of_____ **common decent**

174. There have been a event between the Ediacaran and Early Cambrian faunas.___ __ _____ _____ **Neoproterozoic**

175. Humans arose Ma. and fossil evidence points to repeated human migrations out of Africa?_ _____ _____**6-8**

176. The centrales are type of Diatoms prefers _____ environment? _____ **marine**

177. The Precambrian is divided into eras.____ _____**all**

178. Shelled organisms lived in fresh water have .... than marine water___**Low calcification**

179. Insects first appeared during the ….and diversified rapidly?_____**Early Devonian**

180. A number of groups of microfossils change color with changing_ _____**temperature**

181. In . Francis Crick suggested that RNA was the first genetic molecule._____**1968**

182. The is the most diverse and mast threatened ecosystems?_____ _____**Corals reef**

183. 150KM crater, 10 KM Diameter of Meteorite was identified in…. in Mexico._____**1991**

184. Complex organic molecules arose gradually on a pre-existing, replication platform._____ _____**non-organic**

185. A__. isa lithologically distinct layer within a member or formation and is the smallest recognizable stratigraphic unit._____ _____**bed**

186. The Phylum…. can be traced back to at least the Late Precambrian. when Kimberella probably fed on algae in Ediacaran communities?_____ _____**Mollusca**

187. The….. is characterized by a huge variety of shell shapes, dentitions and muscle scars in mollusks?____ _____**Bivalves**

188. Hadean Eon spans from the origin of the Earth. 4.57 to about__ Ga_____**4**

189. The remains of the activity of ancient organisms. such as burrows and track are termed as ___ fossils._____ _____**trace**

190. From Latin scientia. meaning _____**knowledge**

191. The principle of cross-cutting relationships:_____ **If a body or discontinuity cuts across a stratum, it must have formed after that stratum.**

192. When large animals feed on dead plant or animal tissues, the process is termed - _____ **Scavenging**

193. The Subphylum Crustaceans have classes?_____ _____**Six**

194. In event. the removal of majority of brachipods like spiriferides and the last strophomenides occur?_____ _____**End-Triassic**

195. The explosion in metazoans generated a range of new body plans during a relatively short time interval?_____ _____**Cambrian**

196. The is the most greatest complexity of any of the spiralians?_____**Cephalopods**

197. The Arthropods are also called?_____**Ecdysozne**

198. was the first to picture evolution as a great branching tree.___ _____**Darwin**

199. The asteichthyans. or bony fishes, also appeared in the a_____**devonian**

200. The first metazoans were multicellular with one main cell type called _____**Choanocytes**

201. Molecular clock gives the concept of_____ _____**time estimation**

202. Graptolites (1 myr) and Ammonites (25 kyr) are the best known and most reliable zone _____**macrofossils**

203. established the simple fact that older racks are overlain by younger racks if the sequence has not been inverted law of superposition of strata._____ _____**Nicolaus Steno**

204. The composition and structure of fossil communities._____**paleasynecology**

205. Ostracodes are crustacean arthropods. abundant and widespread in aquatic environments. Appeared first during the ae_____ _____**Early Cambrian**

206. gave the idea of paradigm shift._____ _____**Thomas Kuhn**

207. ____ are the beasts that live in or on the seabed._____ ____**benthos**

208. The …..biota consisted of enigmatic tubular and frond-shaped, mostly sessile organisms that lived during the 635-542 Mya_____ _____**Ediacaran**

209. Driven to extinction in the ___ century by overhunting. the dodo is now an image of human thoughtlessness_____**17th**

210. Phylogenetically uninformative characters fall into ….main categories._____**2**

211. The Jelly fish belong to the class. _____**Scyphozos**

212. Certain developmental abnormalities called ….show former stages of evolution, such as human babies with small tails or excessive hair._____ ___**atavisms**

213. In porifera, …..have most common where a series of sycon chambers access a large central paragaster?_____**Leucon**

214. A membrane-bound structure composed mainly of lipids that self-replicates called -_____ _____**protocell**

215. …..in evolution is when features, or organisms, evolve to look the same perhaps because they live the same way._____ _____**Convergence**

216. The sponges are skeletal organisms. Skeletons are composed of a ?_____**Spongin**

217. __ and …..(1992) gives techniques to unstrain Cambrian trilobites from Kashmir._____ _____**Hughes and Jell**

218. Infaunal organisms live…. the sediment-water interface._____ _____**beneath**

219. Reef-building animals, brachiopods, echinoderms, astracodes trilobites died out during…. mass extinction,__ _____**End-Ordovician**

220. The Phylum ….are usually unsegmented, soft-badied animals with a bady plan based on four features?_____ _____**Mollusca**

221. The Sub phylum Myriapods first appeared during the _____**Mid Silurian**

222. …was linked to the ides of evolution_____**Progressionism**

223. The Earth today can be divided into main provinces of bio-geagraphy._____**six**

224. The Parazoan body plan characterized by cells organized in two layers separated by jelly-like material punctuated by so-called _____**Wandering calls or amosbacytes**

225. The group includes various algae, the coccolithophores and the diatoms._ ___**Chromistans**

226. Platydendron from the Middle Cambrian Burgess Shale has been ascribed to the__ _____ _____**Platyhelminthes**

227. Trilobites have provided key evidence in studies of its_____**Macroevolution**

228. The first conodonts were identified by the Latvian embryalogist and paleontologist Christian Pander in_____**1856**

229. By the Late Carboniferous a very diverse…. fauna had evolved. with farms such as the dragonflies and mayflies capable of powered flight?___ _____**Insect**

230. Entirely soft-bodied organisms. such as …..and , are only preserved in rare cases_____ **both**

231. Many animals such as the soft-badied mollusks (slugs) possess a in which the movement of fluid provides support?_____**Hydreulle Skeleton**

232. The most advanced mollusks. with a head, senses and a nervous system. Nautiloids. Ammonaids and the Colecids fall in -_____**Cephalopods**

233. Humans are more closely related to_____ _____**Chimpanzee**

234. The is simplest of the higher metazoans with a radial diploblastic body plan and stinging cells or cnidoblasts?_____**Cnidarian**

235. The Phylum Hemichordata have classes?_____ _____Two
236. ….helps in identifying the levels of thermal maturity of racks.____ _____Fossils
237. The Tetrapads arose from… during the Devonian?_____ _____fishes
238. In Cnidarians, the stage which is sessile are called?_____ _____Polyps
239. In late Devonian, event like climate change cause removal of families?_____ _____Anypides.& Pememendes
240. The Class Bivalves evolved rapidly in the to include basal forms of all bivalve infrasubclasses? _____ _Early Ordovician
241. The Class Bivalves evolved from ?_____Restroconchs
242. A ….mass extinction event occur at, or close to, the Triassic- Jurassic boundary, 200 Ma__ _____marine
243. A is … naturally occurring assemblage of plants and animals that live in the same place at the same time and regularly interbreed._____ _____population
244. The Bryozoans have fossil species?_____16000
245. describes the biotic and morphological changes along phylogenetic lineages -_____Phillips
246. The Phylum has most common marine animals today which occupy wide range of habitats even at abyssal depths of the oceans?_____ _____Mollusca
247.

For final term exam

QUIZ NO 2

1. John Maynard Smith and Edrs Szathméry (1997) identified…. major steps from the origin of life to human societies with language._____8
2. many of the mammals of North America, such as bears, deer, squirrels, hedgehogs and moles, were found also in_____ _____Eurasia
3. ……is the study of the distribution of species and ecosystems in geographic space and through geological time_____ _____Biogeography
4. Prokeryotes lack a…. and have (usually) 6 single circular chromosome._____ _____nucleus
5. A taxon or two related taxa are found on either side of a barrier to their spread is known as_____dispersalism
6. According to . distribution of plants was closely linked to climate__ _____botanist
7. ….soils suggest that larger plants and animals had moved onto land._____ _____Ordovician
8. …had noted the relationship between the area of an island and the number of species that it contains__ _____all
9. The species diversity increasing by a factor of___ for every doubling of island area_____10
10. The concepts of ecological biogeography. botanical regions and island biogeography had. then. all been recognized by the end of the century._____18th
11. The species 0 is found south of the River Congo and is hence separated from other chimpanzees by a formidable barrier to movement._____ _____bonobo
12. proposed the life forms of plants._____ _____Christen Raunkiser
13. The first tree sized plants appeared in Late period._____ _____Devonian
14. Which one of the following is second major diapsid clade _____ Lepidosaurs
15. The fossile may be classified according to the which of the following mode of behaviour represented _____All of given above
16. A network of collagen fiber on which needle like crystals of hydroxyapatite a form of (apatite capo4) accumulate _____ Bone
17. Sphenodon the tuatara of _____New zeland

18. Lepidosaurus are_____ _____Lizard
19. Monotremes are restricted to _____Australasia
20. Homo sapiens mean _____wise person
21. Human arose ---------- million years ago _____6-8
22. Apes arose from the ---------- _____old word monkey
23. ---------- live in aferica from 2.4 to 1.5ma _____Homo habilis
24. Since 1861 ---------- more skeleton have come to light _____ Nine(9)
25. Archeopteryx is ---------- fossils of ancient dinosaurs _____Bird like
26. Proficient flyer diapsiad were called _____Pterosaurs
27. Synapsid radiated dramatically in the ------- as a new clade , the therapsida. _Late Permian
28. Oldest known bird fossil were found in _____Upper Jurassic
29. The basal vertebrate, laked jaws, jaws probably evolved during the---------- ____Ordovician
30. ---------- more complex, with cones, bars and blades _____Euconodonts
31. Ancestor of human and chimps seprated about _____8-6 ma
32. The membrane which enclose the embryo is called _____Amnion
33. Early bony fishes almost certainly had _____both lungs and gills
34. After the Late Triassic extinction a new group of ….dinosaurs, the sauropodomorphs, radiated dramatically, some like *Plateosaurus.* _____ herbivorous
35. Dinosaurs famously ruled the Earth for 160 Myr of the Mesozoic. Replaced by the mammals 65 Ma.
36. The oldest substantial skeletons, of *Praeanthropus afarensis*, _____3.2 Ma
37. Trace fossils may be classified according to the mode of behavior represented:_____ _____ all of these
38. Plants evolved on land in the _____Ordovician and Silurian,
39. Gorillas seem to have diverged first, about 10 Ma, and the ancestors of humans and chimps separated about _____8–6 Ma.
40. Arnold Ortmann published work based on the distribution of_____crustaceans
41. A species is found on either side of the equatorial regions, but notwithin them is a phenomenon of ._____ _____bipolarity
42. presented the theory of continental drift._____ _____Wegener
43. Croizat published a book named based on the idea of dispersal.____ __Panbiogeography
44. Driven by great processes — the biological process of evolution and the geological process of plate tectonics._____ _____two
45. George Simpson, Karl Schmidt, George Myers who worked on fishes._____ _freshwater
46. The linked concepts of succession and climax, first developed by_____ ____Henry Cowles
47. To understand underlying patterns of distribution of organisms on our planet,…. of species must be known._____ _____nature
48. High mountain chains, expanses of water or areas of arid desert may confine a species…. to region._____ _____ _____particular
49. The tracks of many.. belonging to a wide variety of organisms,could be combined to form a generalized track that connected different regions of the world._____ _____taxa
50. …..of the continents and islands of the Southern Hemisphere, and had suggested that these might be explained partly by the dispersal of floating seeds._____ _____Flora
51. In the early the first landscape- scale ecosystem was subjected to monitoring and manipulative management at Hubbard Brook, a forested mountainside in New Hampshire._____1960s
52. Edward Forbes: (1856) published depth zones and faunal provinces along the coasts of the continents._____5 &25
53. Geologists were able to provide a series of palaeogeographic maps that showed, from the..Period onward, the changing patterns of association of the various tectonic plates_Silurian
54. Larger islands contain a number of individuals, and a greater diversity of species, than smaller islands._____greater

55. All… subspecies lie to the north or east of this river Congo._____chimpanzee
56. British geologist argued that many lines of evidence suggested that the Earth must be many millions of years old._____ _____Lyell
57. A taxon or two related taxa are found on either side of a barrier to their spread is known as ._____dispersalism
58. Raymond Lindemann, in flow in nature._____1942
59. New Guinea contributes only 3% of the world's land area, it contains some 10% of its species of……. organism._____ _____terrestrial
60. Philip Sclaterwas a believed that all species had been created within the area in which they are found today._____ _____ornithologist
61. Global diversification pattern is comprises of mathematical patterns._____8
62. Wegener's theory was renamed as…. in 1960._____pate techoies
63. Xerophytes tolerate levels of moisture._____ _____low
64. Archaeopteryx, found in Upper Jurassic sediments in southern Germany in _____1861
65. Large brainand _are the distinguishing characters of Humans._____ _____bipedalism
66. Second major marine reptile group is_____ _____Plesiosaurs
67. According to exponential model, limitations slow the rate of increase of life.___bath a and b
68. A unique small-sized human species, arose about 13,000 years ago._____h.fl oresiensis
69. Truly modern humans .H sapiens may have arisen as much as -------- and certainly by ------ in aferica _____-400,000 years ago ,150,000 years ago
70. Prokaryotes lack a and have (usually) a single circular chromosome._____ _____nucleus
71. A larger brain means higher intelligence and so greater evolutionary adaptability is an__ _____ __assumption
72. The first person to realize that "similar environments, found in different regions of the world, contained different groupings of organisms is law._____Buffons
73. The extraordinary diversity of tetrapode predators in the sea came to an end _____65Ma
74. Marsupials are included in ------- subclass of mammalia. _____Metatheria
75. A new species, H. erectus, which arose in Africa about___—Ma._____1.9
76. ….had to summate the faunas and floras from every locality within each of the resulting in Peleacontinents._____Palaeobiageagrapher
77. Forster found that Buffon's Law applied to plants as well as to_____ _____Forster
78.

Final term

1. The Spirallians are morphologically diversed clade of protostome animals including all of the following except……..._____Cnidarians
2. Early life was mainly affected by …..…..._____predation
3. Lucy fossil old myb…. _____3.2 million years
4. Reptilian ancestors of mammals had… _____synapsid skull
5. The most advance mollusks are……..._____Cephalopods
6. Amniotes radiated in……..._____Late carboniferous
7. In early corocodilian were most diverse……..._____Jurassic and Cretaceous
8. Crocodile radiate in…..._____jurassic
9. First fossil evidence of bipedalism……. _____Australopithecus afarensis
10. Crocodile more diverse… _____Jurassic and cretaceous
11. Oldest known bird found in……_____Upper Jurassic sediments
12. The oldest human was found is approximately lived _____ago…….._____6Mya

13. German paleontologist ---------- established a classification of trace fossil based on behavior …….**Adolf Seilacher**

14. .are the observation of the appearance and disappearance of species **Patterns**

15. Which Scientist translated Cuvier's idea into English in 1813……**Robert Jameson**

16. ……..argues that all areas connected by one of these tracks had originally formed a single, continuous are that was inhabited by group concerned ……**Panbiogeography**

17. Briggs used pathogen of endamicity of coastal faunas to identify location where there appears to be a zone of unusually rapid faunal change and then used this to distinguish ---------- **zoogeographical regions**

18. Brigg distinguish _____ zoogeographical regions…….**23**

19. The sum of all niche requirements under ideal conditions when the species is given unimpeded access to resources is called……..**Fundamental Niche**

20. ____noted that the range boundaries of Amazonia birds seemed to coincide with the patterns of river…….**Alfred Wallace**

21. Geological study of New Caledonia show that the Island was submerged for long period in the Paleocene……..Eocene so that Island was not available for colonization until about……**37 mya**

22. The ancestors of mammoths and modern Asian elephant probably evolved in……**Africa**

23. During the early Devonian 380mya the fossil record suggest that all the early amphibians and reptile groups evolved in the continents called……..**Euramerica**

24. All of the following part of Gondwana Except……..**North America**

25. Many studies have demonstrated that the richness of the species particularly plant increases with -----**Altitude**

26. Evolution a process by which new species are generated it essentially involve……**Genetic Variation and Subsequent selection**

27. The trade winds found in the both northern and southern hemisphere meet in the region of equators this is known as……..**Intertropical Convergence zone (ITCZ)**

28. _____ is a term sometimes used to describe waves of biological invasions that has occurred in the wake of human invasions by either accidental, transport or deliberate introduction ……**Ecological Imperialism**

29. When two species are attempting to tap a resource in the same way and when that resource in short supply the two are said to be in…….**Competition**

30. The theory of Island biogeography published in…….**1967**

31. All of the following are included in ecological biogeography except……**Altitude**

32. In terms of diversification of life during the early Cambrian _____ was an important part of an ecosystem……**Predation**

33. The focus of human evolution was in…….**Africa**

34. The reptile ancestors of the mammals are……..**synapsid**

35. Chondricthyans is fish……**Cartilaginous**

36. _____is a term sometimes used to describe the wave of biological invasion that has occurred……..**Ecological imperialism**

37. ---------- in on origin of species where devoted to geographical distributions…….**Two Chapters**

38. The oldest substantial skeleton of praeanthropus afarensis lived approximately ---------- ago…….**3.2 Mya**

39. Early vertebrate skeletion…… **cartilage**

40. Certain species play key roles in the ecosystem, and when they are lost there is a sudden drop in the capacity of the ecosystem to function…..**Rivert hypthosis**

41. Paleontology Originate from Greek word "Ontos" which means……**Creature**

42. Today Extenction**…..** **6th**

43. Archaeopetyx is true missing link between……._____**Dinosaurs and Birds**

44. Trilobites dominated assemblages in England and Wales area were described by _____ _____-**William Smith**

45. Fossils in Fold belts are, except……_____.**In Excellent Shape**

46. Where an organism live matters in which of the filters is preservation of the complete fossil record……_____**Ecological Filters**

47. In Climate change in life history of earth has oscillated at least ---------- between icehouse and greenhouse conditions……._____.**5 times**

48. Scala Naturae is ----------_____**The Ladder of life**

49. When local ecological conditions affect the form of an organism during its life time and consequently fossil is called ……._____.**Ecophenotypic Effects**

50. Rohde and Muller (2005) argued that there is____ periodicity in mass extinctions ……_**62 myr**

51. Modification of the Oparin_____Haldane biochemical model is……___**Hydrothermal Model**

52. ___Biomarker indicates that eukaryotes were present around at least 2.7 Ga……._____**Lipid**

53. Chromistans Includes all of the following, except**……..**_____**Protozoa**

54. The Parazoan body plan is seen in……._____**Sponges**

55. Fossils Lagerstattan are assemblages of……____.**inadequate Preserved soft bodied**

56. The Spirallians are morphologically diversed clade of protostome animals including all of the following except……_____**Cinideria**

57. The most advanced mollusks are …….._____**Cephalopods**

58. The "Belly footed" Mollusks are……_____**Gastropods**

59. The acme of the group Gastropoda with the radiation of the siphonal carnivorous neogastropods was in ……._____**Cenozoic**

60. Chelicerates are diversed and heterogeneous which includes…._____ _____ **All of the above (Mites, Scorpions, Spiders)**

61. Water Vascular system is present in……._____.**Echinoderms**

62. Paleontology Originate from Greek word "Ontos" which means_____ _____**Creature**

63. Archaeopetyx is true missing link between____ _____**Dinosaurs and Birds**

64. Fossils in Fold belts are, except__ _____**In Excellent Shape**

65. Where an organism live matters in which of the filters is preservation of the complete fossil record .._____**Ecological Filters**

66. In Climate change in life history of earth has oscillated at least ---------- between icehouse and greenhouse conditions_____**5 times**

67. Scala Naturae is ----------_____**The Ladder of life**

68. When local ecological conditions affect the form of an organism during its life time and consequently fossil is called _____**Ecophenotypic Effects**

69. Rohde and Muller (2005) argued that there is ---------- periodicity in mass extinctions___**62 myr**

70. Modification of the Oparin ---------- Haldane biochemical model is __**Hydrothermal Model**

71. A network of collagen fibers on which needlelike crystals of hydroxyapatite (a form of apatite, CaPO4) accumulate The Skeleton _____**Bone**

72. Cartilage are the type of tissues which are progressively mineralized by: _____**Apatite**

73. The first vertebrates had a _____ skeleton _____**Cartilaginous**

74. The _____ appears in the early embryo as a strip of cells where the backbone will develop Jawless Fishes _____ _____**neural crest**

75. in _____, commonest vertebrates were the conodont animals _____ _____**Late Cambrian and Ordovician**

76. _____ Were jawless, generally armored _____**Ostracoderms**

77. _____ are the extinct genus of jawless fishes that lived in Devonian period. _**Hemicyclaspis**

78. They occur as phosphatic toothlike microfossils, termed elements. _____**Conodonts**

79. Conodonts are divided into ____ groups _____**3**

80. _____more complex, with cones, bars and blades _____ **Euconodonts**
81. The three main morphotypes of conodonts are: _____ **Coniform, Ramiform, Pectiniform**
82. The basal vertebrates, lacked jaws, jaws probably evolved during the_____ ___ **Ordovician**
83. Oldest jawbearing fishes were the_____ _____ **Placoderms**
84. chondrichthyans were the first _____ like cartilaginous fishes _____ **Shark**
85. Small fishes, mostly in the range _____ in length. _____ **50–200 mm**
86. The _____ are Ancestors of most fishes today _____ **Actinopterygians**
87. The first good evidence of tetrapods were _____ long _____ **0.6 and 1 m**
88. _____ are 4000 species of mainly small animals that live in or close to water _____ _____ **Half-way land animals**
89. The oldest fossil form, with reduced limbs, is _____ in age _____ **Jurassic**
90. A part of amniotic egg that collects waste. _____ **Allantois**
91. _____ are Distinguished by the pattern of openings in the side of the skull _____ _____ **Late Carboniferous**
92. The three major skull patterns seen in amniotes:_____ **anapsid,diapsid and synapsid**
93. All modern amniotes produce the ____ egg _____ **Cleidoic**
94. The oldest Turtles and relatives were _____ eaters _____ **small insect**
95. First synapsids were Grouped loosely as "_____" _____ ____ **Pelycosaurs**
96. Reptiles typically have ____ bones in the lower jaw _____ **Six**
97. In mammals, there is a single bone in the lower jaw known as: _____ **the dentary**
98. As compared to the three tiny bones in human ear, reptiles have only one, known as: _____ _____ ___ **stirrup or stapes**
99. Dinosaurs famously ruled the Earth for _____ of the Mesozoic _____ **160 Myr**
100. The mammals achieved great diversity and abundance only after the extinction of: _ **Dinosaurs**
101. A group, includes the dinosaurs, pterosaurs, crocodilians and birds. _____ **Archosaurs**
102. _____ has a lightweight body, Narrow hatchet-shaped skull, a long narrow wing _____ _____ **Proficient flapping flyers**
103. First true lizards are: _____ **Lepidosaurs**
104. The theropods and sauropodomorphs share the primitive _____ pattern _____ **reptilian hip**
105. All other dinosaurs share a unique hip pattern in which the pubis has swung back and runs parallel to:_____ **Ischium**
106. fish-shaped animals, adapted to life in the sea _____ **The ichthyosaurs**
107. Marine reptile _____ uses snake like neck for hunting the fish in water. _____ **Plesiosaurs**
108. The plesiosaur, _____ Swam by beating their paddles in a kind of "flying" motion. _____ **Cryptoclidus**
109. The most famous fosils of Oldest known bird were Found in Upper Jurassic sediments in southern Germany in _____ **1861**
110. Examples of Flightless ratites and ancestors of water birds __ **penguins and birds of prey**
111. Small insect eaters in the Late Triassic and Early Jurassic Remained small through most of the: _____ **Mesozoic**
112. A classic example of an adaptive radiation due to development of teeth _____ _____ **Marsupials & Placentals**
113. _____ are early primates _____ **Humans**
114. Homo sapiens means: _____ **wise person**
115. primates have Good _____ vision _____ **binocular**
116. _____ primates radiated in the Paleocene, Eocene and Oligocene _____ **Basal**
117. _____ in Marmosets and spider monkeys are used as extra limbs in swinging through the trees. _____ **Prehensile Tails**
118. The _____ of Southeast Asia are the most primitive living apes_____ **Gibbons**

119. ___ of the DNA of Gorillas, chimpanzees and humans is identical. _____ **>94%**

120. In_____, two French teams announced human fossils that were 6 Ma. ____**2001 and 2002**

121. The famous skeleton of a female P. afarensis from Ethiopia, called ____ by its discoverer _**Lucy**

122. Lucy's brain is small, only ____ for a height of 1–1.2 m _____**415 cm^3**

123. Another human species, H . erectus had Brain size of _____ in a body up to 1.6 m tall _____ _____ _____ _____**830– 1100 cm^3**

124. Homo erectus Truly modern humans, H . sapiens, may have arisen as much as ____ years ago _____**400,000**

125. _____ had large brains (on average, 1400 cm3) _____**Neandertals**

126. _____, which concentrates on macroscopic (visible with the naked eye) plant remains _____**Paleobotany**

127. Palynology is the study of_____. _____**fossil pollen and spores**

128. Trace fossils are named on the basis of_____ _____**shape and ornamentation**

129. The study of trace fossils is often called _____ **Ichnology**

130. structures formed in hard substrates, such as limestone, shells or wood, for the purpose of protection _____**Borings**

131. non-fecal pellets are regurgitation pellets of_____ **birds and reptiles**

132. Two models for the diversification of marine invertebrate life over the past 600 myr of good quality fossil records. _____**The Empirical Model, The Bias Simulation Model** .

133. the plateau appears clearly only in the _____ data compilation._____ _____**family-level**

134. A single equilibrium level is easier to understand in terms of a: __**global equilibrium model**

135. The Modern _____ radiated dramatically over the last 100 myr_____ **"fauna"**

136. _____ means change with improvement. _____**Progress**

137. _____ are observations of the appearance and disappearance of species._____ **Patterns**

138. _____ are hypotheses that seek to explain the patterns. _____**Processes**

139. The replacement of brachiopods by bivalves is a famous example of: _**Biotic replacements**

140. First living organisms' bacteria: _____**(3500 Ma)**

141. Oldest multicellular eukaryotes: reported from rocks _____ in Canada _____**1200 Ma**

142. A process of reproduction in which a new individual arises by the fusion of two sex cells, or gametes, produced by different individuals _____**Sexual reproduction**

143. Composed of many kinds of cells – muscle cells, nerve cells, epithelial cells and so on _____**Differentiated cells**

144. the study of the distribution of species and ecosystems in geographic space and through geological time. _____**Biogeography**

145. _____ are ideal locations because they allow scientists to look at habitats that new invasive species have only recently colonized _____**Islands**

146. He noted many of the mammals of North America, such as bears, deer, squirrels, hedgehogs and moles, were found also in Eurasia • Travelled via Alaska, when climates were warmer _____ _____ _____ **Georges Buffon**

147. the founder of plant geography _____**German Alexander von Humboldt:**

148. Augustin de Candolle of Geneva and Lamarck published a map showing France divided into _____ regions with different ecological conditions. _____ **five floristic**

149. who was the first to classify the world's flora and show the results on maps? _____ _____ **Danish botanist Joakim Schouw**

150. French Revolution of ____ led to a flowering of French science _____**1789**

151. Power of the Church was broken: _____**18th century**

152. _____ of Natural History, became a powerhouse of ideas and debate in Europe._____ _____ **New National Museum**

153. In 1802, _____ suggested that • 'Lower' organisms might also be found earlier in time and that they might gradually change into the 'higher' forms _____**Jean- Baptiste Lamarck**

154. Geoffroy suggested evolutionary homologies and links between such widely different animals as: _____ **fish and cephalopods**

155. Darwin had spent the _____ after his voyage in detailed research on many other areas of biology that provided evidence for evolution _____ **40 years**

156. _____ is the central paradigm of the earth sciences and Theory of evolution is of the biological sciences _____ **Plate tectonics**

157. ___ was the first to make a world map_____ **Engler**

158. The system of plant regions accepted today is very similar to that of:_____ **Engler**

159. "all species had been created within the area in which they are found today". • This was first formalized by: _____ **British ornithologist Philip Sclater**

160. Presented his theory of continental drift in (1912) **Pangaea Alfred Wegener**

161. Croizat published his ideas in the 1950s and 1960s, his major presentation being his 1958 book: _____ **Panbiogeography**

162. Plants in High levels of heat and moisture _____ **Megatherms**

163. Plants in Moderate levels of heat and moisture _____ **Mesotherms**

164. plants in Low levels of heat and moisture _____ **Microtherms**

165. James Dana divided the _____ into several different zones based on mean minimum temperature. _____ **surface waters**

166. _____ in 1856 published Five depth zones and 25 faunal provinces along the coasts of the continents _____ **Edward Forbes**

167. three British zoologists Assembled _____ of the distributions of fishes based on the patterns of distribution of 27 families. _____ **30 maps**

168. Sheer variety and volume of the works published by: _____ **Alfred Wallace Candolle**

169. New Guinea contributes only _____ of the world's land area and some 10% of its species of terrestrial organism _____ **3%**

170. A species that exists as a series of subspecific forms is termed A _____ Species ___ **Polytypic**

171. All chimpanzee subspecies lie to the north or east of this river _____ **River Congo**

172. A _____ may be deterred from crossing an area of grassland; or a marsh organism may fail to travel across dry habitats to reach the next area of wetland. _____ **forest species**

173. _____ that leaves its forest- floor log, is exposed beetles, shrews and insectivorous birds. _____ **Insect**

174. ____ Dive for their food and bring vegetable matter to the surface from greater depths. _ **Coots**

175. The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called _____ **Klepto-parasitism**

176. Patterns of Distribution Species may evolve in one region, spread to other locations and then become extinct in all but a restricted area where it survives. Species restricted in this way are said to be _____ **Endemic**

177. the two major factors influencing the degree of endemism in an area: _ **Isolation, Stability**

178. Australia is indeed rich in endemics, many of which have a long geological history. This kind of 'fossil endemism' is called_____ **Palaeo-endemism**

179. Anything that tends to make it more difficult for a species to live, grow or reproduce in its environment may prove to be a: _____ **limiting factor**

180. Any regular change in physical or chemical conditions through space that creates a sequence of replacement of one species by another, among both animals and plants. _____ **zonation**

181. the predators can always turn to alternative food species if the numbers of their usual prey should be reduced by climatic or other influences. This is termed. _____ **Prey Switching**

182. ____ reduce the rate of growth of populations because of their negative influence on general fitness, survival and fecundity. _____ **Parasites**

183. The removal of certain species could create effects far in excess of what may originally have been expected. Influential species of this kind are known as. _____ **Keystone Species**

184. The loss of a _____ can cause an avalanche of local extinctions. _____**Keystone Species**

185. Microscopic plankton are capable of changing the depth at which they live, depending on conditions, and this can be regarded as a form of:_____ **vertical migration**

186. ____ of a species can be measured by its geographical distribution _____**Success**

187. The term _____ is used for animals that play a particular role in the community, such as plantsucking insects or insectivorous birds _____**guild**

188. To cope with the physical demands of their environment _____**Species adapt**

189. species tolerating one another, or becoming dependent on one another _____**Co- evolution**

190. Pointing out that no two species have quite the same needs.____ **Individualistic concept**

191. _____ of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem_____ **Removal**

192. A possible alternative model is based on the supposition that all species are equally important to: _____**ecosystem function**

193. certain species play key roles in the ecosystem, and when they are lost there is a sudden drop in the capacity of the ecosystem to function. This is sometimes referred to as: _____ _____**The Rivet Hypothesis**

194. ____ is Defined in terms of inertia, or resistance to change. _____**Stability**

195. Woody savannas – trees > 2 m high, cover only _____ of the land surface _____**30–60%**

196. Variations in climate also result from the pattern of movement of _____**air masses**

197. The 'trade winds', found in both the Northern and Southern Hemispheres, meet in the region of the equator, and this is known as the: _____**intertropical convergence zone (ITCZ)**

198. Approximately _____ species of organism have been described _____**1.8 million**

199. Of the bacteria, only about ____ species have so far been described _____**4000**

200. __ have always been most abundant in the low-latitude (tropical) regions.____ **Angiosperms**

201. Greater extinction rates in the higher latitudes is the instability of climatic conditions over the past ____ years _____**2 million**

202. High-latitude organisms have _____ than those from the low latitudes. _____ _____ **broader geographical ranges**

203. General feature of biogeography first pointed out by ____ in the 1970s _____**E.H. Rapoport**

204. Katherine Smith and James Brown examined the diversity of __ as one proceeds deeper __**fish**

205. The Amazon basin contains approximately _____ flowering plant species, whereas equivalent areas in Africa and in South-East Asia contain only about 40,000 each. _____**90 000**

206. All the land surface area of the hotspots together comprise only about ____ of the Earth's terrestrial total _____**1.4%**

207. The idea that continents could fragment and move across the face of the planet was first suggested by the German meteorologist _____ in 1912 _____**Alfred Wegener**

208. The surface of the Earth, known as the lithosphere, is occupied by a number of areas known as: _____**tectonic plates**

209. The _____ move as well as the continents, the study of their movements is known as plate tectonics rather than continental drift _____**ocean floors**

210. The fossil record suggests that all the early amphibian and reptile groups evolved in the continent called _____ **Euramerica**

211. __ are all the result of the break-up and dispersal of Gondwana ____**Southern continents**

212. mainly the result of the appearance and disappearance of land or sea barriers within North America and Eurasia _____**Northern continents**

213. Apart from humans and the aerial bats, only the __ spread naturally as far as Australia ____**rats**

214. Human beings probably arrived _____ years ago and brought the domestic dog (the ancestor of the dingo) about 3500 years ago _____**60 000 to 40 000**

215. Geological studies show that _____ is the smallest (about 1600 km2) surviving fragment of Gondwana _____**New Caledonia**

216. New Zealand has Its total area, _____, makes it far larger than any other Pacific oceanic island _____**270 000 km2**

217.

BEST OF LUCK

MUHAMMAD IMRAN

MID TERM

1. First Fossil Evidence Of Bipedalism?

The oldest clear evidence for bipedalism: human tracks in volcanic ash from Tanzania, 3.75 Ma. • The oldest substantial skeletons, of Praeanthropus afarensis, 3.2 Ma.

2. What Are Biozones In The Context Of Biostratigraphy?

Biozones: The main operational units of a biostratigraphy. Measurements of the stratigraphic ranges of fossils, or assemblages of fossils, form the basis for the definition of biozones. The known range of a zone fossil is the time between its first and last appearances in a specific rock section, • First appearance datum (FAD) • Last appearance datum (LAD). Establishment of biozones • Quantitative techniques to understand the relationships between rock thickness and time, and to make links from locality to locality.

3. What Is Micropaleontology?

Micropaleontology is the branch of paleontology that studies microfossils, or fossils that require the use of a microscope to see the organism, its morphology and its characteristic

4. What Are Helicoplacoids?

Helicoplacus is the earliest well-studied fossil echinoderm. Fossil plates are known from several regions. Complete specimens were found in Lower Cambrian strata of the White Mountains of California. The animal was a cigar-shaped creature up to 7 centimetres long that stood upright on one end.

5. What Are Anatomic Filters?

Organisms are likely to be preserved only if they have hard parts, a skeleton of some kind. • Entirely softbodied organisms, such as worms and jellyfish, are only preserved in rare cases.

6. Name The "Big Five" Mass Extinctions Event?

The —big fivell Phanerozoic mass extinctions occurred in the • End-Ordovician • Late Devonian • EndPermian • End-Triassic • End-Cretaceous.

7. Write Four Feature Characteristics Of Phylum Mollusks?

Unsegmented soft body with bilateral symmetry.
Presence of an internal or external shell.
A toothed tongue (made mostly of chitin) called the radula.
A mantle which is a fold in the body wall that lines the shell.
Muscular foot (and/or tentacles in some).
(copied from google)

Short Questions (03)

8. Role Of Amateurs?

Amateurs are not paid to work as paleontologists • Collaborations between different sciences is the key.

9. Thermal Maturity?

Helps in identifying the levels of thermal maturity of rocks. {**Thermal maturity** is the extent of heatdriven reactions that alter the composition of organic matter (e.g., conversion of sedimentary organic matter to petroleum or cracking of oil to gas.)}

10. Gaia's Hypothesis?

Gaia hypothesis: Earth as a living organism in equilibrium with its biosphere.

11. Formation of Bed and Member?

A bed, is a lithologically distinct layer within a member or formation and is the smallest recognizable stratigraphic unit. **A member** is a more local lithologic development, usually part of a formation.

12. Isthmus Of Panama?

Barrier for some organisms may provide a corridor for others. • The emergence of the Isthmus of Panama (3 Mya) • Connected North and South America • Separated the Atlantic and Pacific oceans.

13. Types Of Fossils?

Five types of fossils: (a) insect preserved in **amber**, (b) **petrified wood** (**permineralization**), (c) **cast** and**mold** of a clam shell, (d) pyritized ammonite, and (e) compression fossil of a fern. **(Google)** Body fossils: the partial or complete remains of plants or animals 2. Trace fossils: the remains of the activity of ancient organisms, such as burrows and tracks. **(PPT)**

14. Diff scavenging and Decay?

When large animals feed on dead plant or animal tissues, the process is termed **scavenging** • When microbes, such as fungi or bacteria, transform tissues of the dead organism, the process is termed **decay.**

15. Diff Volatile and Refractories?

Soft parts of animals are made from volatiles, forms of carbon that break down readily. • Refractories are the organic carbons much less liable to break down, such as cellulose.

16. Infaunal and Epifaunal?

Infaunal organisms live beneath the sediment–water interface • **Epifaunal** organisms live above it The sediment–water interface

17. Allometry and Isometry?

Allometry is a well-known study, particularly in statistical shape analysis for its theoretical developments, as well as in **biology** for practical applications to the differential growth rates of the parts of a living organism's body.

The word 'allometric' comes from the word for different measure in Greek. Most organisms grow allometrically. We then learned that **isometric** growth is growth in which, the proportions of the organism grow the same.This comes from the Greek word that means same measure. Very few organisms grow isometrically.

18. Endosymbiotic Theory?

Margulis: 1970s. • Prokaryote consumed, or was invaded by, prokaryotes • Two species evolved together in a mutually beneficial way.

Endosymbiotic theory is an evolutionary **theory**that assumes **endosymbiosis** between prokaryotic**endosymbionts** and eukaryotic host cell is the**means** by which organelles such as mitochondria and chloroplasts occur within eukaryotic cells.

19. Insects and Corals?

Insects first appeared during the Early Devonian and diversified rapidly • There are probably 10 million species of living insects. Insects had probably already evolved flight before the Mid Carboniferous, when giant dragonflies patrolled the forests.

Corals Most diverse and most threatened ecosystems • The coral reef • Shallow-water coral reefs in zone extending 30° north and south of equator • 30m or 18 °C.

Long Questions (05)

20. Usefulness of Microfossils Record?

Studying **microfossils** has a lot of **uses**. **Microfossils** are **used to** determine how old a piece of rock is and determine if there is gas or oil in the area. They are also**used to** see what kinds of major geological events took place such as earthquakes or major weather changes such as ice storms.

21. Origin Of Life In Early Precambrian World?

The Precambrian is divided into • Hadean eon • Archaean eon • Proterozoic eon • Hadean Eon spans from the origin of the Earth, 4.57 to about 4 Ga. Beginning of the Hadean • Temperatures on surface were too high • The crust was too unstable for any form of carbonbased life to exist. As the Earth's surface cooled, the lithosphere, its rocky crust, began to differentiate as a cooler upper layer above the underlying asthenosphere. The oldest rocks are from Canada and are dated at 3.8–4 Ga, and some mineral grains from Australia have even been dated to 4.4 Ga Lasted from about 4 to 2.5 Ga. • The oldest sedimentary rocks from the Isua Group in Greenland, dated at 3.8–3.7 Ga. The Proterozoic Eon, from 2.5 Ga to 542 Ma • 2.4 Ga, atm oxygen levels rose to onehundredth or onetenth of modern levels.

around 0.8–0.6 Ga is indicated by increased levels of marine sulfate • The two rises in oxygen levels, at the beginning and end of the Proterozoic.

22. Protozoan Characteristics?

Protozoa are notable for their ability to move independently, a **characteristic** found in the majority of species. They usually lack the capability for photosynthesis, although the genus Euglena is renowned for motility as well as photosynthesis (and is therefore considered both an alga and a**protozoan**).

23. Note On Cambrian Explosion?

The Cambrian explosion generated a range of new body plans during a relatively short time interval.

24. Note On Cephalopodes?

A cephalopod is any member of the molluscan class Cephalopoda such as a squid, octopus or nautilus. These exclusively marine animals are characterized by bilateral body symmetry, a prominent head, and a set of arms or tentacles modified from the primitive molluscan foot.

25. Arthropods Features?

However, the members of this phylum, despite their incredible diversity and sheer numbers, share a number of important distinguishing characteristics.

- Exoskeleton. ...
- Segmented Bodies. ...
- Jointed Appendages. ...
- Bilateral Symmetry. ...
- Open Circulatory System.

26. Echinoderm Water vascular System?

The **water vascular system** is a hydraulic **system**used by **echinoderms**, such as sea stars and sea urchins, for locomotion, food and waste transportation, and respiration. The **system** is composed of canals connecting numerous tube feet.

27. Any 2 Models about Origin of Life?

Scientific models: Some of them now rejected by the evidence • Others still available as potentially valid hypotheses: 1. Spontaneous generation 2. Inorganic model. 3. Extraterrestrial origins 4. Biochemical model 5. Hydrothermal model.

The inorganic model: Complex organic molecules arose gradually on a preexisting, nonorganic replication platform • Silicate crystals • Graham CairnsSmith of Glasgow University, 2007 • not conclusive.

28. Define Paleontology?

Paleontology. [pā′lē-ŏn-tŏl′ə-jē] The scientific study of life in the geologic past, especially through the study of animal and plant fossils.

29. Define Speciation. Two Forms of Speciation?

Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously interbreeding population. Evolution takes place both within • Species lineages (phyletic gradualism) • at the time of speciation (punctuated equilibrium).

30. Among Polychaetes, Why Paddle Worms Have Complete Record?

Polychaetes: A diverse polychaete fauna has been described from the Burgess Shale • Even contains Canada spinosa • Similar to some living polychaetes. **Paddle worms:** most complete fossil record • Record is enhanced by the relatively common preservation of elements of the phosphatic jaw: scolecodonts.

31. Island Geography?

Islands are areas of land that are not connected to a continent and are surrounded by water.

Small **islands** are sometimes called cays, keys, or islets. A group of**islands** is often called an archipelago. There are two main type of **islands**; continental **islands** and oceanic **islands**.

32. Zones Of Fossil?

Index **fossils** (also known as guide **fossils** or indicator**fossils**) are **fossils** used to **define** and identify geologic periods (or faunal stages). Index **fossils** must have a short vertical range, wide geographic distribution and rapid evolutionary trends.

33. Bangiomorpha?

Bangiomorpha pubescens is a red alga. It is the first known sexually reproducing organism. A multicellular fossil of **Bangiomorpha** pubescens was recovered from the Hunting Formation in Somerset Island, Canada that strongly resembles the modern red alga Bangia despite occurring in rocks dating to 1,047 million years ago.

34. Oldest Rocks and Their Age?

The oldest rocks are from Canada and are dated at 3.8–4 Ga, and some mineral grains from Australia have even been dated to 4.4 Ga.

35. Nanoplankton?

Like other phytoplankton, **nanophytoplankton** are microscopic organisms that obtain energy through the process of photosynthesis and must therefore live in the upper sunlit layer of ocean or other bodies of water. ... The term**nanophytoplankton** is derived from the far more widely used term**nannoplankton**/**nanoplankton**.

36. Sexual Dimorphism?

Sexual dimorphism is the condition where the two sexes of the same species exhibit different characteristics beyond the differences in their **sexual**organs. The condition occurs in many animals and some plants.

37. Deutrostomes?

Deuterostomia is a subtaxon of the Bilateria branch of the subkingdom Eumetazoa, within Animalia, and are distinguished from protostomes by their deuterostomic embryonic development; in **deuterostomes**, the first opening (the blastopore) becomes the anus, while in protostomes, it becomes the mouth.

38. Trilobitomorpha?

Highly derived arthropods • lack specialized mouthparts • Tagmata comprising the cephalon, thorax and pygidium • Trilobitomorph appendages. Mainly the trilobites and over 15,000 species are known • Common: throughout the Paleozoic • Extinction: end of the Permian.

39. Island Biogeography?

Insular **biogeography** or **island biogeography** is a field within **biogeography** that examines the factors that affect the species richness of isolated natural communities. The theory was originally developed to explain species richness of actual **islands**, principally oceanic.

40. What Are Plesiomorphies?

In phylogenetics, a **plesiomorphy**, symplesiomorphy or symplesiomorphic character is an ancestral character (trait state) shared by two or more taxa - but also with other taxa linked earlier in the clade (that is, having an earlier last common ancestor, with them, than theirs).

41. Biostratigraphy?

The branch of stratigraphy concerned with fossils and their use in dating rock formations.

42. Biozones Range?

Biostratigraphic unit or **biozones** are intervals of geological strata that are **defined** on the basis of their characteristic fossil taxa. ... The International Commission on Stratigraphy defines the following types of **Biozones**: **Range** zones, subdivided into taxon-**range**zones and concurrent-**range** zones.

43. Triploblastic Body Plan?

Few basic body plans have appeared in the fossil record • most animals have triploblastic architecture • Three fundamental body layers.

44. Oldest fossil n their ages?

The oldest eukaryote fossil may be Grypania, • A coiled, spaghetti-like organism that has been reported from rocks as old as 1.85 Ga
The oldest locomotory trace fossils are from about 550 Ma northwest Russia • Fecal strings from rocks some 600 Ma suggesting Bacteria in rocks up to 3.2 Ga, indicated by stromatolites.

45. Write the name of animal groups who hits by end permian mass extinction event?

The end-Permian mass extinction was the largest of all time • Probably caused massive volcanic eruptions, leading to acid rain and global anoxia**. (ppts)**
During the period a number of animal groups became extinct, including the trilobites, tabulate and rugose corals, and blastoids (see mass extinction). **Amphibians** and **reptiles** continued to be the dominant land animals and gymnosperms replaced ferns, clubmosses, and horsetails as the dominant **plants**. "Permian. **(google)**

46. What are the Trilobitomorpha? what are their different body parts? 3

Highly derived arthropods
- lack specialized mouthparts
- Tagmata comprising the cephalon, thorax and pygidium
- Trilobitomorph appendages

Mainly the trilobites and over 15,000 species are known
- Common: throughout the Paleozoic
- Extinction: end of the Permian

47. What is "Great Ordovician Biodiversification Event" (GOBE)?

Great Ordovician biodiversification event (GOBE), • Evolutionary radiation of animal life during the Ordovician Period, 40 my after the CE. Followed a series of Cambrian– Ordovician Extinction events • Fauna went on to dominate the Palaeozoic • Marine diversity increased Ordovician diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels.

Important Definitions (02)

Burgess Shale:

The Burgess Shale is a fossil-bearing deposit exposed in the Canadian Rockies of British Columbia, Canada. It is famous for the exceptional preservation of the soft parts of its fossils. At 508 million years old, it is one of the earliest fossil beds containing soft-part imprints

Lagerstatten:

A Lagerstätte (German: [ˈlaːgɐˌʃtɛtə], from Lager 'storage, lair' Stätte 'place'; plural **Lagerstätten**) is a sedimentary deposit that exhibits extraordinary fossils with exceptional preservation—sometimes including preserved soft tissues.

Lithostratigraphy:

Lithostratigraphy is a sub-discipline of stratigraphy, the geological science associated with the study of strata or rock layers. Major focuses include geochronology, comparative geology, and petrology. In general a stratum will be primarily igneous or sedimentary relating to how the rock was formed.

Signor-Lipps Effect:

The **Signor–Lipps effect** is a paleontological principle proposed by Philip W. **Signor** and Jere H. **Lipps**which states that, since the fossil record of organisms is never complete, neither the first nor the last organism in a given taxon will be recorded as a fossil.

Chronostratigraphy:

Chronostratigraphy is the branch of stratigraphy that studies the age of rock strata in relation to time. The ultimate aim of **chronostratigraphy** is to arrange the sequence of deposition and the time of deposition of all rocks within a geological region, and eventually, the entire geologic record of the Earth.

Tophonomy:

Taphonomy is the study of how organisms decay and become fossilized.

Coenosis:

An ecosystem, originally **defined** by Tansley (1935), is a biotic community (or biocenosis) along with its physical environment (or biotope). In ecological studies, biocenosis is the emphasis on relationships between species in an area.

Variation and Speciation:

Variation, in **biology**, any difference between cells, individual organisms, or groups of organisms of any species caused either by genetic differences (genotypic **variation**) or by the effect of environmental factors on the expression of the genetic potentials (phenotypic **variation**). **Speciation** is the evolutionary process by which populations evolve to become distinct species. ... There are four

geographic modes of **speciation** in nature, based on the extent to which speciating populations are isolated from one another: allopatric, peripatric, parapatric, and sympatric.

Cladistics:

Cladistics is an approach to biological classification in which organisms are categorized in groups based on the most recent common ancestor.

Fidelity:

The **fidelity** of a DNA polymerase is the result of accurate replication of a desired template. ... High-**fidelity** PCR utilizes DNA polymerases that couple low misincorporation rates with proofreading activity to give faithful replication of the target DNA of interest.

Five big mass extinction event?

The —big five‖ Phanerozoic mass extinctions occurred in the • End-Ordovician • Late Devonian • EndPermian • End-Triassic • End-Cretaceous.

What is Bangiomorpha?

One of the oldest multicellular organisms is Bangiomorpha • 1.2 Ga • Multicellular and a member of a modern group that engages in sex.

Fossil zone?

Index fossils (also known as guide fossils or indicatorfossils) are fossils used to define and identify geologic periods (or faunal stages). Index fossils must have a short vertical range, wide geographic distribution and rapid evolutionary trends. (**Internet**)

Nanoplankton?

Very small unicellular plankton, at the limits of resolution of light microscopy.

What is paleontology?

Paleontology or palaeontology is the scientific study of life that existed prior to, and sometimes including, the start of the Holocene Epoch. It includes the study of fossils to determine organisms' evolution and interactions with each other and their environments.

Speciation and write two model? / What is Speciation?

Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously interbreeding population.

Evolution takes place both within • Species lineages (phyletic gradualism) • at the time of speciation (punctuated equilibrium). The evolution of life may be represented by a single branching phylogenetic tree. • Cladistics is a method of reconstructing phylogeny based on the identification of shared derived characters (homologies).

Molecular sequencing provides additional evidence for reconstructing and dating the tree of life. • DNA has been extracted from fossils such as woolly mammoths, but not from truly ancient fossils

What is micropaleontology?

Micropaleontology is the branch of paleontology that studies microfossils, or fossils that require the use of a microscope to see the organism, its morphology and its characteristic details.

Old rock and their ages? / Which is the most oldest rock on earth write age?

The oldest rocks are from Canada and are dated at 3.8–4 Ga, and some mineral grains from Australia have even been dated to 4.4 Ga.

Define biostarigraphy?

The branch of stratigraphy concerned with fossils and their use in dating rock formations.

What are known range of biozone?

Biostratigraphic unit or biozones are intervals of geological strata that are defined on the basis of their characteristic fossil taxa. ... The International Commission on Stratigraphy defines the following types of Biozones: Range zones, subdivided into taxon-rangezones and concurrent-range zones.

Four characteristic phylum mollusca?

Unsegmented soft body with bilateral symmetry. Presence of an internal or external shell.

A toothed tongue (made mostly of chitin) called the radula.

A mantle which is a fold in the body wall that lines the shell.

Muscular foot (and/or tentacles in some). **(copied from google)**

Sexual dimorphisim?

Sexual dimorphism is the condition where the two sexes of the same species exhibit different characteristics beyond the differences in their sexualorgans. The condition occurs in many animals and some plants.

Animals groups hit by end Permian name?

The end-Permian mass extinction was the largest of all time • Probably caused massive volcanic eruptions, leading to acid rain and global anoxia. **(ppts)**

During the period a number of animal groups became extinct, including the trilobites, tabulate and rugose corals, and blastoids (see mass extinction). Amphibians and reptiles continued to be the dominant land animals and gymnosperms replaced ferns, clubmosses, and horsetails as the dominant plants. "Permian. **(google)**

What is biogeography?

The branch of biology that deals with the geographical distribution of plants and animals.

Duetrostomes?

Deuterostomia is a subtaxon of the Bilateria branch of the subkingdom Eumetazoa, within Animalia, and are distinguished from protostomes by their deuterostomic embryonic development; in deuterostomes, the first opening (the blastopore) becomes the anus, while in protostomes, it becomes the mouth.

Define induction?

Sir Francis Bacon • The enquirer might hope to see common patterns among the observations, and these common patterns would point to an explanation, or law of nature.

Why the polychates have incomplete fossil? While paddle worm have complete fossil record?

Polychaetes: A diverse polychaete fauna has been described from the Burgess Shale • Even contains Canada spinosa • Similar to some living polychaetes. Paddle worms: most complete fossil record • Record is enhanced by the relatively common preservation of elements of the phosphatic jaw: scolecodonts.

Define dimorphic organism with example?

Dimorphic fungi are fungi that can exist in the form of both mold and yeast. Anexample is Penicillium marneffei, a human pathogen that grows as a mold at room temperature, and as a yeast at human body temperature.

Anatomic filters?

Organisms are likely to be preserved only if they have hard parts, a skeleton of some kind. • Entirely softbodied organisms, such as worms and jellyfish, are only preserved in rare cases.

Island biography?

Islands are areas of land that are not connected to a continent and are surrounded by water. Small islands are sometimes called cays, keys, or islets. A group ofislands is often called an archipelago. There are two main type of islands; continental islands and oceanic islands.

What is great Ordovician biodiversity event (GOBE)?

Great Ordovician biodiversification event (GOBE), • Evolutionary radiation of animal life during the Ordovician Period, 40 my after the CE. Followed a series of Cambrian– Ordovician Extinction events • Fauna went on to dominate the Palaeozoic • Marine diversity increased Ordovician diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels.

Questions:

island biogeography...2

Island biogeography is the study of the species composition and species richness on islands

nannoplankton..2

Defined as plankton less than 63 µm across, the smallest standard mesh size for sieves • Includes organicwalled and siliceous forms

.sexual dimorphism...3

dimorphism, in which males and females may show different sizes, and different specialized features variations within species.

Growth stages, where there may be quite different larval and adult stages, or where body form alters during Variations Within the Species.
Ecophenotypic effects • Where local ecological conditions affect the form of an organism during its lifetime

.oldest fossil n their ages..5

The oldest eukaryote fossil may be Grypania, • A coiled, spaghetti-like organism that has been reported from rocks as old as 1.85 Ga

The oldest locomotory trace fossils are from about 550 Ma northwest Russia • Fecal strings from rocks some 600 Ma suggesting

Bacteria in rocks up to 3.2 Ga, indicated by stromatolites

.features of mollusca..5

The Phylum Mollusca can be traced back to at least the Late Precambrian, when Ki m b e r ell a

probably fed on algae in Ediacaran communities

Early Mollusks

Characterized by some short-lived, unusual forms • Molluskan features: mantle, mineralized shell and radula • Members of the small shelly fauna

Triploblastic body plan 3 marks

Seen in most other animal • Has three layers of tissues from the outside in: the ectoderm, mesoderm and endoderm.

Where on earth the oldest rock are formed? What are their ages. 5 marks

The oldest rocks are from Canada and are dated at 3.8–4 Ga, and some mineral grains from Australia have even been dated to 4.4 Ga

The archaeaneon: Lasted from about 4 to 2.5 Ga. • The oldest sedimentary rocks from the Isua Group in Greenland, dated at 3.8–3.7 Ga. **6**

Deuterostomes 5 marks

“Second mouth” however, have a mouth arising from a secondary opening • True blastopore develops as anus • Echinoderm– Hemichordate Chordate (ppts)

Deuterostomia, (Greek: “second mouth”), group of animals—including those of the**phyla Echinodermata** (e.g., **starfish**, **sea urchins**), **Chordata** (e.g., **sea squirts**,**lancelets**, and **vertebrates**), **Chaetognatha** (e.g., arrowworms), and **Brachiopoda**(e.g., lamp shells)—classified together on the basis of embryological development (google)

What is paleontology(2)

The branch of science concerned with fossil animals and plants. • the study of fossils to determine organisms' evolution & ecology

How anatomic filters helps to preserve complete fossil record(3)

Organisms are likely to be preserved only if they have hard parts, a skeleton of some kind. • Entirely soft-bodied organisms, such as worms and jellyfish, are only preserved in rare cases.

What are the four characteristics of phylum Mollusca(5)

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A mantle which is a fold in the body wall that lines the shell.

Muscular foot (and/or tentacles in some).

What are the big five mass extinction events(5)

During mass extinctions, 20–90% of species were wiped out; these include a broad range of organisms, and the events appear to have happened rapidly

It is difficult to study mass extinctions in the Precambrian • There have been a Neoproterozoic event between the Ediacaran and Early Cambrian faunas.

The “big five” Phanerozoic mass extinctions occurred in the

• End-Ordovician

• Late Devonian

• End-Permian

• End-Triassic

End-Cretaceous who introduce the concept of induction and explain it 2

Sir Francis Bacon • The enquirer might hope to see common patterns among the observations, and these common patterns would point to an explanation, or law of nature. Induction **2): define speciation and write name of two models2**

Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously interbreeding population.

“The process of splitting of a population to form two species is speciation ”.

• The allopatric • (“other homeland”) model

• The geographic model (based on the establishment of geographic barriers

:write the name of animal groups who hits by end permian mass extinction event.3

The end-Permian mass extinction was the largest of all time • Probably caused massive volcanic eruptions, leading to acid rain and global anoxia.(ppts)

During the period a number of animal groups became extinct, including the trilobites, tabulate and rugose corals, and blastoids (see mass extinction). **Amphibians** and **reptiles** continued to be the dominant land animals and gymnosperms replaced ferns, clubmosses, and horsetails as the dominant **plants**. "Permian. (google)

What is micropaleontology? 2

A multidisciplinary science, focused on the study of microorganisms or the microscopic parts of larger organisms

What are the Trilobitomorpha? what are their different body parts? 3

Highly derived arthropods

• lack specialized mouthparts

• Tagmata comprising the cephalon, thorax and pygidium

• Trilobitomorph appendages

Mainly the trilobites and over 15,000 species are known

• Common: throughout the Paleozoic

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Great Ordovician biodiversification event (GOBE), • Evolutionary radiation of animal life during the Ordovician Period, 40 my after the CE.

Followed a series of Cambrian– Ordovician Extinction events • Fauna went on to dominate the Palaeozoic • Marine diversity increased

Ordovician diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels.

What is biostratigraphy 2m

Biostratigraphy is the establishment of fossil based successions and their use in stratigraphic correlation • Biozones: The main operational units of a biostratigraphy

What is the known range of zone fossils 3m

The known range of a zone fossil is the time between its first and last appearances in a specific rock section, • First appearance datum (FAD) • Last appearance datum (LAD) **in polycheate, why the worms has a complete fossil record 3m**

Record is enhanced by the relatively common preservation of elements of the phosphatic jaw scolecodonts. The earthworm, which has been around for **600 million years**, is spread across the earth and provides a basic biological model for many other species including humans came out top

What is micropalaeontology?

Micropaleontology is a multidisciplinary science, focused on the study of microorganisms or the microscopic parts of larger organisms.

What is island biogeography?

Island biogeography is the study of the species composition and species richness on islands.

What is location and age of the oldest rock?

The oldest rocks are from Canada and are dated at 3.8–4 Ga, and some mineral grains from Australia have even been dated to 4.4 Ga.

What is GOBE?

Great Ordovician biodiversification event (GOBE).

Why do the paddle worm have complete fossil record?

Record is enhanced by the relatively common preservation of elements of the phosphatic jaw: scolecodonts

What is Biogeography?

Island biogeography is the study of the species composition and species richness on islands.(google search)

What is speciation

" The process of splitting of a population to form two species is speciation ".

Define induction?

Sir Francis Bacon • The enquirer might hope to see common patterns among the observations, and these common patterns would point to an explanation, or law of nature.

Define dimorphic organism with example?

dimorphism, in which males and females may show different sizes, and different specialized features (horns,)

FINAL TERM

1. Biogeography of South America: 10

South America

The history of the biota of South America has been dominated by the effects of the great earth-engine of plate tectonics. Some of these effects were caused by the westward drift of the continent, which led to mountain building and consequent climatic changes, but most were the result of changes in its relationship with North America. As noted in the description of the origin of the Caribbean islands in the 'The West Indies' section, there had been islands in that region ever since the Americas started to move westward in the Early Cretaceous. This may have been the route by which some North American dinosaurs dispersed southward in the Mid-Cretaceous, accompanied by some early types of marsupial and placental mammal. But this connection has always been tenuous, sometimes permitting passage and sometimes becoming broken. This led to cycles of immigration, isolation and evolution, and final new immigration and extinction.

The Early Cenozoic:

The mammal fauna of South America today is characterized by a few marsupials (opossums) and a diversity of edentate placental mammals known as the Xenarthra (sloths, armadillos and South American anteaters) that are hardly known elsewhere. However, in the Late Cretaceous, South America had a mammal fauna of strange marsupials and of ungulate placentals quite unlike anything known elsewhere in the world. Alfred Wallace, one of the founders of zoogeography noted that the range boundaries of Amazonian birds seemed to coincide with the pattern of the rivers. It is fascinating that recent research shows that closely related taxa of some birds are to be found on either side of rivers throughout Amazonia.

2. What are three types of radiated amniotes 5

Amniotes radiated during the Late Carboniferous, giving rise to three main clades. These are distinguished by the pattern of openings in the side of the skull, especially the **temporal openings** behind the eye socket. These three skull patterns diagnose the key clades among amniotes which are

i. **The anapsid** ("no arch") is the primitive state skull pattern, since there are no temporal openings. The Anapsida include various early forms such as *Hylonomus*, as well as some Permian and Triassic reptiles, and the turtles. The two other skull patterns seen in amniotes are

ii. **The synapsid** ("same arch"), where there is a lower temporal opening. The Synapsida include the "mammal-like reptiles" and the mammals.

iii. **The diapsid** ("two arch") pattern, where there are two temporal openings. The Diapsida includes a number of early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and birds. .

3. When modern birds appear?5

Modern groups of birds appeared in the latest Cretaceous and Early Tertiary, including flightless ratites and ancestors of water birds, penguins and birds of prey. The perching birds or songbirds, consisting of 5000 species today, radiated in the Miocene

4. What is Kleptoparasitisim with example? 5

Kleptoparasitism (literally, parasitism by theft) is a form of feeding in which one animal takes prey or other food from another that has caught, collected, or otherwise prepared the food, including stored food (as in the case of cuckoo bees, which lay their eggs on the pollen masses made by other bees; food resources could also be in the form of hosts of parasitic or parasitoid wasps). The term is also used to describe the stealing of nest material or other inanimate objects from one animal by another. One very widespread duck in both continents is the gadwall. It feeds mainly on submerged vegetation, which is most abundant in nutrient- rich (eutrophic) waters. It cannot dive, so it feeds by upending, and this generally restricts it to the shallow parts of a lake, where it may encounter some competition from other water birds, such as the mallard. Coots (both Eurasian coot and the American coot dive for their food and bring vegetable matter to the surface from greater depths. The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called **kleptoparasitism**, and it is an effective way of widening the niche of the gadwall.

5. Write a note on geographical history of New Caledonia 5

New Caledonia

Geological studies show that New Caledonia is the smallest (about 1600 km2) surviving fragment of Gondwana, having rifted away from Australia in the Late Cretaceous 80–65 mya and reaching its present position 1500 km east of it 50 mya. It is a hotspot for diversity: 77% of its flowering plant species are endemic to the island, as are 98% of the species of the flowering plant family Sapotaceae.

In view of the large amount of research that shows the frequency of long- distance dispersal in such scenarios, most biogeographers today believe that this was the mechanism of arrival of the ancestors of the **New Caledonian taxa.**

Geological studies show that the island was submerged for long periods in the **Paleocene– Eocene**, so that the island was not available for colonization until about 37 mya. The New Caledonian species must therefore have diverged from their ancestors after this date.

Panbiogeographers, however, **reject long**- **distance dispersal**. They therefore have to theorize that the Cretaceous ancestors of the present biota were widespread across the Pacific islands, including on supposedly formerly exposed large plateaux such as the Ontong Java plateau.

Biogeographers in Sweden and France have investigated the history of the New Caledonian species of the angiosperm family Sapotaceae. They found that these resulted from nine separate colonizations of the island between 33 and 4.2 mya, dates that are about 40 million years after the rifting of New Caledonia from Australia. New Guinea and Australia were the most important sources of the arrivals, those from Pacific islands arriving only more recently, 27–24 mya. These results are clearly totally incompatible with the panbiogeographers' theories.

6. History of old world tropical mammal fauna of Africa 5

Africa contains the best known and most distinctive of the Old World tropical mammal fauna. Molecular studies of the interrelationships of the different orders of placentals suggest that the common ancestor of a group of African placentals entered the continent in the Late Cretaceous. This group is made up of the elephants, hyracoids (conies), aquatic sirenians (sea cows), elephant shrews, aardvarks, Cape golden moles, insectivorous tenrecs and extinct embrithopods; they have been placed together in a super order, Afrotheria, that is endemic to Africa.

The early horse *Hipparion*, which had evolved in North America, appeared in both Eurasia and Africa, while rhinoceros, hyenas and sabre- toothed cats dispersed from Africa to Eurasia. The elephants originated in Africa in the Eocene, migrated into Eurasia in the Early Miocene and migrated back later in the Miocene. The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into Eurasia 2 million years later.

7. Three amniotic layers in embryo and their functions.

- Enclose the embryo (the amnion)
- Collect waste (the allantois)
- line the eggshell (the chorion)
- Yolk, a yellow material rich in protein

8. difference between Sympatry AND Allopatry .

Sympatry

An initially interbreeding population that splits into two or more distinct species sharing a common range exemplifies sympatric speciation. Darwin's finches and their beak sizes

Allopatric speciation

Also referred to as geographic speciation, is a mode of speciation that occurs when biological populations of the same species become isolated from each other.

9. Redundant species hypothesis? 3

According to the *redundant species hypothesis*, the removal of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem. Here, the function of an ecosystem could refer to its overall productivity, its rate of nutrient cycling or its general self- sustainability.

10. What is meant by Biogeography? 2

Biogeography is the study of the distribution of species and ecosystems in geographic space and through geological time.

11. What is meant by Niche? 3

The demands that an organism places on its environment in terms of physical and chemical conditions, space and food supply help to define what ecologists call its **niche**. But the concept of the niche goes beyond the basic physics and chemistry of its habitat and covers all aspects of how the organism makes a living. It includes the food an animal requires, but also encompasses the way in which it acquires that food. The kestrel is a bird that hunts small mammals by day, whereas an owl performs a similar activity by night. Swallows catch aerial insects by day, and bats have the same feeding strategy at night.

12. Name the old amniotes and where it was found?

The oldest-known amniote, *Hylonomus* from the mid-Carboniferous of Canada is known only from its skeleton; no amniotic eggs are known from the Carboniferous.

13. What was the brain capacity of Homo habilis? 3

The first species, *H. habilis,* lived in Africa from 2.4 to 1.5 Ma, and had a brain capacity of 630–700 cm3 in a body only 1.3 m tall. *H. habilis* may have used tools.

14. Describe the Island biogeography of New Candolle 3

Another early contributor was Candolle, who pointed out that the age, climate and degree of isolation of an island, and whether or not it was volcanic, would also affect the diversity of its flora. Nevertheless, the sheer variety and volume of the works on island biogeography published by Alfred Wallace mark him as the real founder of studies on this subject.

15. Trace fossils excrement names, how they are different to other? 3

***Fecal pellets and fecal strings*:** small pellets, usually less than 10 mm in length, or strings of excrement.
***Coprolites*:** discrete fecal masses, usually more than 10 mm in length, and usually the product of vertebrates.

16. Describe the work of James Dana, and Edward Forbes in the Biogeography of Ocean 5

American scientist **James Dana**, who later became an eminent geologist. His brief paper, published in 1853, divided the surface waters of the globe into several different zones based on their mean minimum temperature.

Three years later 1856, the British zoologist **Edward Forbes** published the first comprehensive work, recognizing five depth zones and 25 faunal provinces along the coasts of the continents. He was the first to recognize the enormous Indo- Pacific faunal region.

17. Zoogeography 2

Developing from early 19th century onward with a different emphasis Birds and Mammals. Do not show a close correlation to local ecology. Early Zoogeographer: Prichard in 1826 and Swainson in 1835. Six regions. This was first formalized in 1858 by the British ornithologist *Philip Sclater*. He believed that all species had been created within the area in which they are found today.

18. Traces on bedding planes

- **Tracks**: sets of discrete footprints, usually formed by arthropods or vertebrates
- **Trails:** continuous traces, usually formed by the whole body, either trave

Excrement

- **Fecal pellets and fecal strings**: small pellets, usually <10 mm in length, or strings of excrement
- **Coprolites**: discrete fecal masses, usually >10 mm in length, and usually the product of vertebrates

Naming

- Given formal names, often based on Latin and Greek, just like living and fossil plants and animals
- Trace fossil genera: ichnogenera
- Trace fossil species: ichnospecies

19. Adaptive Radiations

One of the classic observations of large-scale evolution. A radiation is when a clade expands relatively rapidly. The adjective “adaptive”: happening because of some particular adaptation in the clade.

20: What are two key defining characters of Jawless fishes? 2

Two key defining characters: The head and neural crest tissues.

The neural crest appears in the early embryo as a strip of cells lying just below the outer skin, the ectoderm, of the embryo, above the line of where the backbone will develop.

21.What are characteristic features of Pterosaurs? 3

The pterosaurs were proficient flapping flyers, with a lightweight body, narrow hatchetshaped skull and a long narrow wing supported on a spectacularly elongated fourth finger of the hand. Pterosaurs were covered with hair, and were almost certainly **endothermic**. Most pterosaurs fed on fishes caught in coastal seas, but others were insectivorous.

22. What is Study of trace fossil is called? 2

The study of trace fossils, often called ichnology (from the Greek ichnos, a trace).

23. Functions of Geographic Information System.

Modern biogeography often employs the use of Geographic Information Systems (GIS), to understand the factors affecting organism distribution, and to predict future trends in organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

24. Work of Leon Croizat in field of Biogeography.

He amassed a vast array of distributional data, representing each biogeographical pattern as a line, or track, connecting its known areas of distribution. Croizat studied the distribution patterns of many unrelated taxa, and for each he drew lines or ‘tracks’ on the map linking the areas in which they are

found. In many cases, these lines were similar enough in position to be combined as 'generalized tracks', shown here. He found that the tracks of many taxa, belonging to a wide variety of organisms, could be combined to form a generalized track that connected different regions of the world

25. Oldest clear evidence for bipedalism

The oldest clear evidence for bipedalism was the find of some human tracks in volcanic ash from Tanzania, dated at 3.75 Ma. The oldest substantial skeletons, of *Praeanthropus afarensis*, come from rocks dated at about 3.2 Ma and also show clear anatomic evidence for advanced bipedalism, but still an ape-sized brain.

26. Name the most famous fossils. 3

One of the most famous fossils is *Archaeopteryx*, the oldest known bird. The first specimen was found in Upper Jurassic sediments in southern Germany in 1861, and was hailed as the ideal "missing link" or proof of evolution in action. Here was an animal with a beak, wings and feathers, so it was clearly a bird, but it still had a reptilian bony tail, claws on the hand, and teeth. Since 1861, nine more skeletons have come to light, the last two in 1992 and 2005.

27. What is the Biotic Replacement, give example?

Biotic Replacements are an obvious feature of the history of life. These are times when one group of plants or animals replaces another. The replacement of brachiopods by bivalves is a famous example. This had always been seen as a progressive process: the common view is that brachiopods are less adaptable than bivalves, and they clearly succumbed to long-term competition, perhaps lasting for tens or hundreds of millions of years.

28. The work of Albert Gunther, and Arbold Ortman in Biogeography 3

In 1880, the British zoologist **Albert Gunther** published a book on fishes in which he recognized 10 different regions in the distribution of shore fishes.
The German **Arnold Ortmann** published a similar work based on the distribution of crustaceans such as crabs and lobsters.

29.Erythrosachus

Small and large meat eaters such as *Erythrosuchus* appeared, one of the first of the archosaurs, a group that was later to include the dinosaurs, pterosaurs, crocodilians and birds.

30. Modern Birds

Appeared in the latest Cretaceous and Early Tertiary. Flightless ratites and ancestors of water birds, penguins and birds of prey. The perching birds radiated in the Miocene.

31.Biotic Replacements

- An obvious feature of the history of life.
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- of brachiopods by bivalves is a famous example

32.Homo habilis

Lived in Africa from 2.4 to 1.5 Ma. Brain capacity of 630– 700 cm3 in a body only 1.3 m tall.

33.Jawless Fishes

Two key defining characters: the head and neural crest tissues

34.Redundant Species Hypothesis

Removal of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem.

35.Archaeopteryx

Archaeopteryx is the most famous fossils. Archaeopteryx is Oldest known bird found in Upper Jurassic sediments in southern Germany in 1861.

36.Traces on Bedding Planes

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37.Biogeography

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To predict future trends and understand the factors affecting organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

39.Kleptoparasitism

A very widespread duck in the gadwall (Anas strepera). Coots (both Eurasian coot, Fulica atra, and the American coot, Fulica americana). Dive for their food and bring vegetable matter to the surface from greater depths. The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called kleptoparasitism, and it is an effective way of widening the niche of the gadwall.

40.Oldest Amniote

Oldest-known amniote, *Hylonomus* from the mid-Carboniferous of Canada. Only skeleton; no amniotic eggs are known from the Carboniferous. E.g The earliest reptile, The mid-Carboniferous Reptile *Hylonomus*

41.Radiated amniotes

Amniotes: Late Carboniferous Three main clades distinguished by the pattern of openings in the side of the skull. Temporal openings behind the eye socket.

The three major skull patterns seen in amniotes: anapsid, diapsid and synapsid. (Based on Carroll 1987.)

- **The anapsid** (“no arch”) skull pattern, no temporal openings.
- **The synapsid** (“same arch”), lower temporal opening,
- **The diapsid** (“two arch”) pattern, two temporal openings.

Three Clades

- **The Anapsida**: early forms such as *Hylonomus*, as well as some Permian and Triassic reptiles, and the turtles.
- **The Synapsida**: “mammal-like reptiles” and the mammals
- **The Diapsida** : Early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and birds. All modern amniotes produce the cleidoic egg

42.Amnionic layers of embryo of reptiles and their function

- Enclose the embryo (the amnion) Collect waste (the
- allantois) line the eggshell (the chorion) Yolk, a yellow
- material rich in protein.
-

43.Marine biogeography scientist works

- **James Dana:** (1853), divided the surface waters into several different zones based on mean minimum temperature.

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45.Niche:

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In view of the large amount of research that shows the frequency of long_distance dispersal in such scenarios, most biogeographers today believe that this was the mechanism of arrival of the ancestors of the New Caledonian taxa. Geological studies show that the island was submerged for long periods in the Paleocene–Eocene, so that the island was not available for colonization until about 37 mya. The New Caledonian species must therefore have diverged from their ancestors after this date

- Panbiogeographers reject long_distance dispersal.They theorize that the Cretaceous ancestors of the present biota were widespread across the Pacific islands, including on supposedly formerly exposed large plateaux such as the Ontong Java plateau
- Biogeographers have investigated the history of the angiosperm family Sapotaceae. Resulted from nine colonizations of the island between 33 and 4.2 mya, dates that are about 40 million years after the rifting of New Caledonia from Australia.

New Guinea and Australia were the most important sources of the arrivals, those from Pacific islands arriving only more recently, 27–24 mya. These results are clearly totally incompatible with the panbiogeographers' theories.

49.Old world tropic fauna of Africa

- Contains the best known & most distinctive of Old World tropical mammal fauna. Molecular studies of the interrelationships of the different orders of placentals suggest that the common ancestor of a group of African placentals entered the continent in the Late Cretaceous.
- This group is made up of the elephants, hyracoids (conies), aquatic sirenians (sea cows), elephant shrews, aardvarks, Cape golden moles, insectivorous tenrecs and extinct embrithopods; they have been placed together in a superorder, Afrotheria that is endemic to Africa.
- Much of the continent was covered by rainforests of the tropical ever wet and subtropical summer wet biomes for much of the early Cenozoic, until the Late Miocene. About 19 mya in the Miocene, a firm land connection with Eurasia took place in what is now the Middle East
- The seaway was finally broken near the beginning of the Late Miocene, 12 mya, by rising mountains in Arabia, Turkey and the Middle East. It was at this time that the early horse Hipparion, which had evolved in North
  America, appeared in both Eurasia and Africa
- The elephant originated in Africa in the Eocene, migrated into Eurasia in the Early Miocene and migrated back later in the Miocene The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into Eurasia 2 million years later.

50.write the old history mammals of fauna of africa?

- The Old World monkeys of Africa arose from Eocene The apes Radiated
- in Africa in the Miocene.
- The apes spread out from Africa into the Middle East, Asia and southern Europe by the mid-Miocene
- A number of incomplete human fossils reported during the 1990s from Africa, dated between 6 and 4 Ma.
- Homo habilis Lived in Africa from 2.4 to 1.5 Ma
- A new species, H. erectus, which arose 1.9 Ma in Africa
- Truly modern humans, H. sapiens, may have arisen as much as 400,000 years ago, and certainly by 150,000 years ago, in Africa,

51.Biogeographic history of South America

- The history of the biota of South America has been dominated by the effects of the great earth_engine of plate tectonics. Westward drift of the continent, mountain building and consequent climatic changes changes in its relationship with North America.

- There had been islands in that region ever since the Americas started to move westward in the Early Cretaceous. This may have been the route by which some North American dinosaurs dispersed southward in the
  Mid_Cretaceous, with some early types of marsupial and placental mammal.

- But this connection has always been tenuous, sometimes permitting passage and sometimes becoming broken. This led to cycles of immigration, isolation and evolution, and final new immigration and extinction. The mammal fauna of South America today is characterized by a few marsupials (opossums) and a diversity of edentate placental mammals known as the Xenarthra (sloths, armadillos and South American anteaters) that are hardly known elsewhere.

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BEST OF LUCK